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Methods and user interfaces for handling user requests

Abstract

The present disclosure generally relates to methods and user interfaces for managing user requests for specific operations. In some embodiments, a computer system, after receiving a request of a first type performs a requested operation and, when notification criteria are met, initiates display of a notification on an external electronic device. In some embodiments, a computer system displays a configuration user interface that includes user-specific examples of requests and/or responses to requests.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims priority to U.S. Provisional Patent Application No. 63/105,805, entitled “METHODS AND USER INTERFACES FOR HANDLING USER REQUESTS”, filed Oct. 26, 2020, the entire contents of which is hereby incorporated by reference.

FIELD
(1) The present disclosure relates generally to computer user interfaces, and more specifically to methods and user interfaces for managing user requests for specific operations.
BACKGROUND
(2) The number of electronic devices, and particularly smart devices, in users' homes continues to increase. These devices are required to perform increasingly complex tasks, including responding to user requests for specific operations.
BRIEF SUMMARY
(3) Some techniques for managing user requests for specific operations using electronic devices, however, are generally cumbersome and inefficient. For example, some existing techniques use a complex and time-consuming user interface, which may include multiple key presses, keystrokes, and/or voice inputs. Some existing techniques do not sufficiently inform users that certain types of operations have been requested, based on certain criteria (e.g., user-defined notification criteria), thereby reducing security and/or potentially compromising privacy, when the operations relate to personal information. Some existing techniques also do not effectively leverage the interconnections between devices to optimize request performance and user notifications. Existing techniques require more time than necessary, wasting user time and device energy. This latter consideration is particularly important in battery-operated devices.
(4) Accordingly, the present technique provides electronic devices with faster, more efficient methods and interfaces for responding to user requests for specific operations. Such methods and interfaces optionally complement or replace other methods for responding to user requests for specific operations. Such methods and interfaces reduce the cognitive burden on a user and produce a more efficient human-machine interface. For battery-operated computing devices, such methods and interfaces conserve power and increase the time between battery charges. For requests for operations that relate to personal and/or confidential data, such methods and interfaces can increase security and enhance privacy.
(5) In accordance with some embodiments, a method, performed at a computer system that is in communication with one or more input devices and a first external electronic device is described. The method includes: receiving, via the one or more input devices, a first request, of a first type, to perform a first operation; and in response to receiving the first request of the first type: in

accordance with a determination that a first set of notification criteria are met: performing the first operation; and initiating a process to display a notification on the first external electronic device; and in accordance with a determination that the first set of notification criteria are not met, performing the first operation without initiating the process to display the notification on the first external electronic device.
(6) In accordance with some embodiments, a non-transitory computer-readable storage medium storing one or more programs configured to be executed by one or more processors of a computer system that is in communication with one or more input devices and a first external device is described. The one or more programs including instructions for: receiving, via the one or more input devices, a first request, of a first type, to perform a first operation; and in response to receiving the first request of the first type: in accordance with a determination that a first set of notification criteria are met: performing the first operation; and initiating a process to display a notification on the first external electronic device; and in accordance with a determination that the first set of notification criteria are not met, performing the first operation without initiating the process to display the notification on the first external electronic device.
(7) In accordance with some embodiments, a transitory computer-readable storage medium storing one or more programs configured to be executed by one or more processors of a computer system that is in communication with one or more input devices and a first external device is described. The one or more programs including instructions for: receiving, via the one or more input devices, a first request, of a first type, to perform a first operation; and in response to receiving the first request of the first type: in accordance with a determination that a first set of notification criteria are met: performing the first operation; and initiating a process to display a notification on the first external electronic device; and in accordance with a determination that the first set of notification criteria are not met, performing the first operation without initiating the process to display the notification on the first external electronic device.
(8) In accordance with some embodiments, a computer system, comprising: one or more input devices; one or more processors; and memory storing one or more programs configured to be executed by the one or more processors is described. The one or more programs including instructions for: receiving, via the one or more input devices, a first request, of a first type, to perform a first operation; and in response to receiving the first request of the first type: in accordance with a determination that a first set of notification criteria are met: performing the first operation; and initiating a process to display a notification on the first external electronic device; and in accordance with a determination that the first set of notification criteria are not met, performing the first operation without initiating the process to display the notification on the first external electronic device.
(9) In accordance with some embodiments, a computer system is described. The computer system comprising: one or more input devices means for receiving, via the one or more input devices, a first request, of a first type, to perform a first operation; and means for, in response to receiving the first request of the first type: in accordance with a determination that a first set of notification criteria are met: performing the first operation; and initiating a process to display a notification on the first external electronic device; and in accordance with a determination that the first set of notification criteria are not met, performing the first operation without initiating the process to display the notification on the first external electronic device.
(10) In accordance with some embodiments, a method performed at a computer system that is in communication with a display generation component is described. The method includes: displaying, via the display generation component, a first configuration user interface associated with a process for configuring data request operations for a respective user, wherein the first configuration user interface includes a set of one or more indications, personalized for the respective user, that includes a first indication, wherein: in accordance with a determination that the respective user is a first user, the first indication includes a representation of first data that is

associated with the first user; and in accordance with a determination that the respective user is a second user, different from the first user, the first indication includes a representation of second data, different from the first data, that is associated with the second user.

(11) In accordance with some embodiments, a non-transitory computer-readable storage medium storing one or more programs configured to be executed by one or more processors of a computer system that is in communication with a display generation component is described. The one or more programs including instructions for: displaying, via the display generation component, a first configuration user interface associated with a process for configuring data request operations for a respective user, wherein the first configuration user interface includes a set of one or more indications, personalized for the respective user, that includes a first indication, wherein: in accordance with a determination that the respective user is a first user, the first indication includes a representation of first data that is associated with the first user; and in accordance with a determination that the respective user is a second user, different from the first user, the first indication includes a representation of second data, different from the first data, that is associated with the second user.

(12) In accordance with some embodiments, a transitory computer-readable storage medium storing one or more programs configured to be executed by one or more processors of a computer system that is in communication with a display generation component is described. The one or more programs including instructions for: displaying, via the display generation component, a first configuration user interface associated with a process for configuring data request operations for a respective user, wherein the first configuration user interface includes a set of one or more indications, personalized for the respective user, that includes a first indication, wherein: in accordance with a determination that the respective user is a first user, the first indication includes a representation of first data that is associated with the first user; and in accordance with a determination that the respective user is a second user, different from the first user, the first indication includes a representation of second data, different from the first data, that is associated with the second user.

(13) In accordance with some embodiments, a computer system, comprising: a display generation component; one or more processors; and memory storing one or more programs configured to be executed by the one or more processors is described. The one or more programs including instructions for: displaying, via the display generation component, a first configuration user interface associated with a process for configuring data request operations for a respective user, wherein the first configuration user interface includes a set of one or more indications, personalized for the respective user, that includes a first indication, wherein: in accordance with a determination that the respective user is a first user, the first indication includes a representation of first data that is associated with the first user; and in accordance with a determination that the respective user is a second user, different from the first user, the first indication includes a representation of second data, different from the first data, that is associated with the second user.

(14) In accordance with some embodiments, a computer system is described. The computer system comprising: a display generation component; means for displaying, via the display generation component, a first configuration user interface associated with a process for configuring data request operations for a respective user, wherein the first configuration user interface includes a set of one or more indications, personalized for the respective user, that includes a first indication, wherein: in accordance with a determination that the respective user is a first user, the first indication includes a representation of first data that is associated with the first user; and in accordance with a determination that the respective user is a second user, different from the first user, the first indication includes a representation of second data, different from the first data, that is associated with the second user.

(15) Executable instructions for performing these functions are, optionally, included in a non-transitory computer-readable storage medium or other computer program product configured for

execution by one or more processors. Executable instructions for performing these functions are, optionally, included in a transitory computer-readable storage medium or other computer program product configured for execution by one or more processors.
(16) Thus, devices are provided with faster, more efficient methods and interfaces for responding to user requests for specific operations, thereby increasing the effectiveness, efficiency, and user satisfaction with such devices. Such methods and interfaces may complement or replace other methods for responding to user requests for specific operations.

Description

DESCRIPTION OF THE FIGURES
(1) For a better understanding of the various described embodiments, reference should be made to the Description of Embodiments below, in conjunction with the following drawings in which like reference numerals refer to corresponding parts throughout the figures.
(2) FIG. **1**A is a block diagram illustrating a portable multifunction device with a touch-sensitive display in accordance with some embodiments.
(3) FIG. **1**B is a block diagram illustrating exemplary components for event handling in accordance with some embodiments.
(4) FIG. **2** illustrates a portable multifunction device having a touch screen in accordance with some embodiments.
(5) FIG. **3** is a block diagram of an exemplary multifunction device with a display and a touch-sensitive surface in accordance with some embodiments.
(6) FIG. **4**A illustrates an exemplary user interface for a menu of applications on a portable multifunction device in accordance with some embodiments.
(7) FIG. **4**B illustrates an exemplary user interface for a multifunction device with a touch-sensitive surface that is separate from the display in accordance with some embodiments.
(8) FIG. **5**A illustrates a personal electronic device in accordance with some embodiments.
(9) FIG. **5**B is a block diagram illustrating a personal electronic device in accordance with some embodiments.
(10) FIG. **5**C illustrates an electronic device in accordance with some embodiments.
(11) FIG. **5**D is a block diagram illustrating an electronic device in accordance with some embodiments.
(12) FIGS. **6**A-**6**V illustrate exemplary user interfaces for managing user requests for specific operations and scenarios exemplifying the same.
(13) FIG. **7** is a flow diagram illustrating a method for managing user requests for specific operations, in accordance with some embodiments.
(14) FIGS. **8**A-**8**D illustrate exemplary user interfaces for configuring management of user requests for specific operations.
(15) FIG. **9** is a flow diagram illustrating a method for configuring management of user requests for specific operations, in accordance with some embodiments.
DESCRIPTION OF EMBODIMENTS
(16) The following description sets forth exemplary methods, parameters, and the like. It should be recognized, however, that such description is not intended as a limitation on the scope of the present disclosure but is instead provided as a description of exemplary embodiments.
(17) There is a need for electronic devices that provide efficient methods and interfaces for responding to user requests for specific operations. In particular, there is a need for devices and methods and provide users with options for managing notifications, using the interconnection between devices, to better inform users that specific operations have been requested. Such techniques can reduce the cognitive burden on a user who makes requests for specific operations

and/or who wish to monitor the occurrence of such requests, thereby enhancing productivity and security. Further, such techniques can reduce processor and battery power otherwise wasted on redundant user inputs.
(18) Below, FIGS. **1**A-**1**B, **2**, **3**, **4**A-**4**B, and **5**A-**5**D provide a description of exemplary devices for performing the techniques for managing user requests for specific operations. FIGS. **6**A-**6**V illustrate exemplary user interfaces for managing user requests for specific operations. FIG. **7** is a flow diagram illustrating methods of managing user requests for specific operations in accordance with some embodiments. The user interfaces and scenarios in FIGS. **6**A-**6**V are used to illustrate the processes described below, including the processes in FIG. **7**. FIGS. **8**A-**8**D illustrate exemplary user interfaces for configuring management of user requests for specific operations. FIG. **9** is a flow diagram illustrating methods of configuring management of user requests for specific operations in accordance with some embodiments. The user interfaces in FIGS. **8**A-**8**D are used to illustrate the processes described below, including the processes in FIG. **9**.
(19) In addition, in methods described herein where one or more steps are contingent upon one or more conditions having been met, it should be understood that the described method can be repeated in multiple repetitions so that over the course of the repetitions all of the conditions upon which steps in the method are contingent have been met in different repetitions of the method. For example, if a method requires performing a first step if a condition is satisfied, and a second step if the condition is not satisfied, then a person of ordinary skill would appreciate that the claimed steps are repeated until the condition has been both satisfied and not satisfied, in no particular order. Thus, a method described with one or more steps that are contingent upon one or more conditions having been met could be rewritten as a method that is repeated until each of the conditions described in the method has been met. This, however, is not required of system or computer readable medium claims where the system or computer readable medium contains instructions for performing the contingent operations based on the satisfaction of the corresponding one or more conditions and thus is capable of determining whether the contingency has or has not been satisfied without explicitly repeating steps of a method until all of the conditions upon which steps in the method are contingent have been met. A person having ordinary skill in the art would also understand that, similar to a method with contingent steps, a system or computer readable storage medium can repeat the steps of a method as many times as are needed to ensure that all of the contingent steps have been performed.
(20) Although the following description uses terms "first," "second," etc. to describe various elements, these elements should not be limited by the terms. These terms are only used to distinguish one element from another. For example, a first touch could be termed a second touch, and, similarly, a second touch could be termed a first touch, without departing from the scope of the various described embodiments. The first touch and the second touch are both touches, but they are not the same touch.
(21) The terminology used in the description of the various described embodiments herein is for the purpose of describing particular embodiments only and is not intended to be limiting. As used in the description of the various described embodiments and the appended claims, the singular forms "a," "an," and "the" are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will also be understood that the term "and/or" as used herein refers to and encompasses any and all possible combinations of one or more of the associated listed items. It will be further understood that the terms "includes," "including," "comprises," and/or "comprising," when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.
(22) The term "if" is, optionally, construed to mean "when" or "upon" or "in response to determining" or "in response to detecting," depending on the context. Similarly, the phrase "if it is determined" or "if [a stated condition or event] is detected" is, optionally, construed to mean "upon

determining" or "in response to determining" or "upon detecting [the stated condition or event]" or "in response to detecting [the stated condition or event]," depending on the context.
(23) Embodiments of electronic devices, user interfaces for such devices, and associated processes for using such devices are described. In some embodiments, the device is a portable communications device, such as a mobile telephone, that also contains other functions, such as PDA and/or music player functions. Exemplary embodiments of portable multifunction devices include, without limitation, the iPhone®, iPod Touch®, and iPad® devices from Apple Inc. of Cupertino, California. Other portable electronic devices, such as laptops or tablet computers with touch-sensitive surfaces (e.g., touch screen displays and/or touchpads), are, optionally, used. It should also be understood that, in some embodiments, the device is not a portable communications device, but is a desktop computer with a touch-sensitive surface (e.g., a touch screen display and/or a touchpad). In some embodiments, the electronic device is a computer system that is in communication (e.g., via wireless communication, via wired communication) with a display generation component. The display generation component is configured to provide visual output, such as display via a CRT display, display via an LED display, or display via image projection. In some embodiments, the display generation component is integrated with the computer system. In some embodiments, the display generation component is separate from the computer system. As used herein, "displaying" content includes causing to display the content (e.g., video data rendered or decoded by display controller **156**) by transmitting, via a wired or wireless connection, data (e.g., image data or video data) to an integrated or external display generation component to visually produce the content.
(24) In the discussion that follows, an electronic device that includes a display and a touch-sensitive surface is described. It should be understood, however, that the electronic device optionally includes one or more other physical user-interface devices, such as a physical keyboard, a mouse, and/or a joystick.
(25) The device typically supports a variety of applications, such as one or more of the following: a drawing application, a presentation application, a word processing application, a website creation application, a disk authoring application, a spreadsheet application, a gaming application, a telephone application, a video conferencing application, an e-mail application, an instant messaging application, a workout support application, a photo management application, a digital camera application, a digital video camera application, a web browsing application, a digital music player application, and/or a digital video player application.
(26) The various applications that are executed on the device optionally use at least one common physical user-interface device, such as the touch-sensitive surface. One or more functions of the touch-sensitive surface as well as corresponding information displayed on the device are, optionally, adjusted and/or varied from one application to the next and/or within a respective application. In this way, a common physical architecture (such as the touch-sensitive surface) of the device optionally supports the variety of applications with user interfaces that are intuitive and transparent to the user.
(27) Attention is now directed toward embodiments of portable devices with touch-sensitive displays. FIG. **1**A is a block diagram illustrating portable multifunction device **100** with touch-sensitive display system **112** in accordance with some embodiments. Touch-sensitive display **112** is sometimes called a "touch screen" for convenience and is sometimes known as or called a "touch-sensitive display system." Device **100** includes memory **102** (which optionally includes one or more computer-readable storage mediums), memory controller **122**, one or more processing units (CPUs) **120**, peripherals interface **118**, RF circuitry **108**, audio circuitry **110**, speaker **111**, microphone **113**, input/output (I/O) subsystem **106**, other input control devices **116**, and external port **124**. Device **100** optionally includes one or more optical sensors **164**. Device **100** optionally includes one or more contact intensity sensors **165** for detecting intensity of contacts on device **100** (e.g., a touch-sensitive surface such as touch-sensitive display system **112** of device **100**). Device

100 optionally includes one or more tactile output generators **167** for generating tactile outputs on device **100** (e.g., generating tactile outputs on a touch-sensitive surface such as touch-sensitive display system **112** of device **100** or touchpad **355** of device **300**). These components optionally communicate over one or more communication buses or signal lines **103**.

(28) As used in the specification and claims, the term “intensity” of a contact on a touch-sensitive surface refers to the force or pressure (force per unit area) of a contact (e.g., a finger contact) on the touch-sensitive surface, or to a substitute (proxy) for the force or pressure of a contact on the touch-sensitive surface. The intensity of a contact has a range of values that includes at least four distinct values and more typically includes hundreds of distinct values (e.g., at least 256). Intensity of a contact is, optionally, determined (or measured) using various approaches and various sensors or combinations of sensors. For example, one or more force sensors underneath or adjacent to the touch-sensitive surface are, optionally, used to measure force at various points on the touch-sensitive surface. In some implementations, force measurements from multiple force sensors are combined (e.g., a weighted average) to determine an estimated force of a contact. Similarly, a pressure-sensitive tip of a stylus is, optionally, used to determine a pressure of the stylus on the touch-sensitive surface. Alternatively, the size of the contact area detected on the touch-sensitive surface and/or changes thereto, the capacitance of the touch-sensitive surface proximate to the contact and/or changes thereto, and/or the resistance of the touch-sensitive surface proximate to the contact and/or changes thereto are, optionally, used as a substitute for the force or pressure of the contact on the touch-sensitive surface. In some implementations, the substitute measurements for contact force or pressure are used directly to determine whether an intensity threshold has been exceeded (e.g., the intensity threshold is described in units corresponding to the substitute measurements). In some implementations, the substitute measurements for contact force or pressure are converted to an estimated force or pressure, and the estimated force or pressure is used to determine whether an intensity threshold has been exceeded (e.g., the intensity threshold is a pressure threshold measured in units of pressure). Using the intensity of a contact as an attribute of a user input allows for user access to additional device functionality that may otherwise not be accessible by the user on a reduced-size device with limited real estate for displaying affordances (e.g., on a touch-sensitive display) and/or receiving user input (e.g., via a touch-sensitive display, a touch-sensitive surface, or a physical/mechanical control such as a knob or a button).

(29) As used in the specification and claims, the term “tactile output” refers to physical displacement of a device relative to a previous position of the device, physical displacement of a component (e.g., a touch-sensitive surface) of a device relative to another component (e.g., housing) of the device, or displacement of the component relative to a center of mass of the device that will be detected by a user with the user's sense of touch. For example, in situations where the device or the component of the device is in contact with a surface of a user that is sensitive to touch (e.g., a finger, palm, or other part of a user's hand), the tactile output generated by the physical displacement will be interpreted by the user as a tactile sensation corresponding to a perceived change in physical characteristics of the device or the component of the device. For example, movement of a touch-sensitive surface (e.g., a touch-sensitive display or trackpad) is, optionally, interpreted by the user as a “down click” or “up click” of a physical actuator button. In some cases, a user will feel a tactile sensation such as an “down click” or “up click” even when there is no movement of a physical actuator button associated with the touch-sensitive surface that is physically pressed (e.g., displaced) by the user's movements. As another example, movement of the touch-sensitive surface is, optionally, interpreted or sensed by the user as “roughness” of the touch-sensitive surface, even when there is no change in smoothness of the touch-sensitive surface. While such interpretations of touch by a user will be subject to the individualized sensory perceptions of the user, there are many sensory perceptions of touch that are common to a large majority of users. Thus, when a tactile output is described as corresponding to a particular sensory perception of a user (e.g., an “up click,” a “down click,” “roughness”), unless otherwise stated, the generated

tactile output corresponds to physical displacement of the device or a component thereof that will generate the described sensory perception for a typical (or average) user.

(30) It should be appreciated that device **100** is only one example of a portable multifunction device, and that device **100** optionally has more or fewer components than shown, optionally combines two or more components, or optionally has a different configuration or arrangement of the components. The various components shown in FIG. **1**A are implemented in hardware, software, or a combination of both hardware and software, including one or more signal processing and/or application-specific integrated circuits.

(31) Memory **102** optionally includes high-speed random access memory and optionally also includes non-volatile memory, such as one or more magnetic disk storage devices, flash memory devices, or other non-volatile solid-state memory devices. Memory controller **122** optionally controls access to memory **102** by other components of device **100**.

(32) Peripherals interface **118** can be used to couple input and output peripherals of the device to CPU **120** and memory **102**. The one or more processors **120** run or execute various software programs (such as computer programs (e.g., including instructions)) and/or sets of instructions stored in memory **102** to perform various functions for device **100** and to process data. In some embodiments, peripherals interface **118**, CPU **120**, and memory controller **122** are, optionally, implemented on a single chip, such as chip **104**. In some other embodiments, they are, optionally, implemented on separate chips.

(33) RF (radio frequency) circuitry **108** receives and sends RF signals, also called electromagnetic signals. RF circuitry **108** converts electrical signals to/from electromagnetic signals and communicates with communications networks and other communications devices via the electromagnetic signals. RF circuitry **108** optionally includes well-known circuitry for performing these functions, including but not limited to an antenna system, an RF transceiver, one or more amplifiers, a tuner, one or more oscillators, a digital signal processor, a CODEC chipset, a subscriber identity module (SIM) card, memory, and so forth. RF circuitry **108** optionally communicates with networks, such as the Internet, also referred to as the World Wide Web (WWW), an intranet and/or a wireless network, such as a cellular telephone network, a wireless local area network (LAN) and/or a metropolitan area network (MAN), and other devices by wireless communication. The RF circuitry **108** optionally includes well-known circuitry for detecting near field communication (NFC) fields, such as by a short-range communication radio. The wireless communication optionally uses any of a plurality of communications standards, protocols, and technologies, including but not limited to Global System for Mobile Communications (GSM), Enhanced Data GSM Environment (EDGE), high-speed downlink packet access (HSDPA), high-speed uplink packet access (HSUPA), Evolution, Data-Only (EV-DO), HSPA, HSPA+, Dual-Cell HSPA (DC-HSPDA), long term evolution (LTE), near field communication (NFC), wideband code division multiple access (W-CDMA), code division multiple access (CDMA), time division multiple access (TDMA), Bluetooth, Bluetooth Low Energy (BTLE), Wireless Fidelity (Wi-Fi) (e.g., IEEE 802.11a, IEEE 802.11b, IEEE 802.11g, IEEE 802.11n, and/or IEEE 802.11ac), voice over Internet Protocol (VoIP), Wi-MAX, a protocol for e-mail (e.g., Internet message access protocol (IMAP) and/or post office protocol (POP)), instant messaging (e.g., extensible messaging and presence protocol (XMPP), Session Initiation Protocol for Instant Messaging and Presence Leveraging Extensions (SIMPLE), Instant Messaging and Presence Service (IMPS)), and/or Short Message Service (SMS), or any other suitable communication protocol, including communication protocols not yet developed as of the filing date of this document.

(34) Audio circuitry **110**, speaker **111**, and microphone **113** provide an audio interface between a user and device **100**. Audio circuitry **110** receives audio data from peripherals interface **118**, converts the audio data to an electrical signal, and transmits the electrical signal to speaker **111**. Speaker **111** converts the electrical signal to human-audible sound waves. Audio circuitry **110** also

receives electrical signals converted by microphone **113** from sound waves. Audio circuitry **110** converts the electrical signal to audio data and transmits the audio data to peripherals interface **118** for processing. Audio data is, optionally, retrieved from and/or transmitted to memory **102** and/or RF circuitry **108** by peripherals interface **118**. In some embodiments, audio circuitry **110** also includes a headset jack (e.g., **212**, FIG. **2**). The headset jack provides an interface between audio circuitry **110** and removable audio input/output peripherals, such as output-only headphones or a headset with both output (e.g., a headphone for one or both ears) and input (e.g., a microphone).

(35) I/O subsystem **106** couples input/output peripherals on device **100**, such as touch screen **112** and other input control devices **116**, to peripherals interface **118**. I/O subsystem **106** optionally includes display controller **156**, optical sensor controller **158**, depth camera controller **169**, intensity sensor controller **159**, haptic feedback controller **161**, and one or more input controllers **160** for other input or control devices. The one or more input controllers **160** receive/send electrical signals from/to other input control devices **116**. The other input control devices **116** optionally include physical buttons (e.g., push buttons, rocker buttons, etc.), dials, slider switches, joysticks, click wheels, and so forth. In some embodiments, input controller(s) **160** are, optionally, coupled to any (or none) of the following: a keyboard, an infrared port, a USB port, and a pointer device such as a mouse. The one or more buttons (e.g., **208**, FIG. **2**) optionally include an up/down button for volume control of speaker **111** and/or microphone **113**. The one or more buttons optionally include a push button (e.g., **206**, FIG. **2**). In some embodiments, the electronic device is a computer system that is in communication (e.g., via wireless communication, via wired communication) with one or more input devices. In some embodiments, the one or more input devices include a touch-sensitive surface (e.g., a trackpad, as part of a touch-sensitive display). In some embodiments, the one or more input devices include one or more camera sensors (e.g., one or more optical sensors **164** and/or one or more depth camera sensors **175**), such as for tracking a user's gestures (e.g., hand gestures) as input. In some embodiments, the one or more input devices are integrated with the computer system. In some embodiments, the one or more input devices are separate from the computer system.

(36) A quick press of the push button optionally disengages a lock of touch screen **112** or optionally begins a process that uses gestures on the touch screen to unlock the device, as described in U.S. patent application Ser. No. 11/322,549, "Unlocking a Device by Performing Gestures on an Unlock Image," filed Dec. 23, 2005, U.S. Pat. No. 7,657,849, which is hereby incorporated by reference in its entirety. A longer press of the push button (e.g., **206**) optionally turns power to device **100** on or off. The functionality of one or more of the buttons are, optionally, user-customizable. Touch screen **112** is used to implement virtual or soft buttons and one or more soft keyboards.

(37) Touch-sensitive display **112** provides an input interface and an output interface between the device and a user. Display controller **156** receives and/or sends electrical signals from/to touch screen **112**. Touch screen **112** displays visual output to the user. The visual output optionally includes graphics, text, icons, video, and any combination thereof (collectively termed "graphics"). In some embodiments, some or all of the visual output optionally corresponds to user-interface objects.

(38) Touch screen **112** has a touch-sensitive surface, sensor, or set of sensors that accepts input from the user based on haptic and/or tactile contact. Touch screen **112** and display controller **156** (along with any associated modules and/or sets of instructions in memory **102**) detect contact (and any movement or breaking of the contact) on touch screen **112** and convert the detected contact into interaction with user-interface objects (e.g., one or more soft keys, icons, web pages, or images) that are displayed on touch screen **112**. In an exemplary embodiment, a point of contact between touch screen **112** and the user corresponds to a finger of the user.

(39) Touch screen **112** optionally uses LCD (liquid crystal display) technology, LPD (light emitting polymer display) technology, or LED (light emitting diode) technology, although other display technologies are used in other embodiments. Touch screen **112** and display controller **156**

optionally detect contact and any movement or breaking thereof using any of a plurality of touch sensing technologies now known or later developed, including but not limited to capacitive, resistive, infrared, and surface acoustic wave technologies, as well as other proximity sensor arrays or other elements for determining one or more points of contact with touch screen **112**. In an exemplary embodiment, projected mutual capacitance sensing technology is used, such as that found in the iPhone® and iPod Touch® from Apple Inc. of Cupertino, California.

(40) A touch-sensitive display in some embodiments of touch screen **112** is, optionally, analogous to the multi-touch sensitive touchpads described in the following U.S. Pat. No. 6,323,846 (Westerman et al.), U.S. Pat. No. 6,570,557 (Westerman et al.), and/or U.S. Pat. No. 6,677,932 (Westerman), and/or U.S. Patent Publication 2002/0015024A1, each of which is hereby incorporated by reference in its entirety. However, touch screen **112** displays visual output from device **100**, whereas touch-sensitive touchpads do not provide visual output.

(41) A touch-sensitive display in some embodiments of touch screen **112** is described in the following applications: (1) U.S. patent application Ser. No. 11/381,313, “Multipoint Touch Surface Controller,” filed May 2, 2006; (2) U.S. patent application Ser. No. 10/840,862, “Multipoint Touchscreen,” filed May 6, 2004; (3) U.S. patent application Ser. No. 10/903,964, “Gestures For Touch Sensitive Input Devices,” filed Jul. 30, 2004; (4) U.S. patent application Ser. No. 11/048,264, “Gestures For Touch Sensitive Input Devices,” filed Jan. 31, 2005; (5) U.S. patent application Ser. No. 11/038,590, “Mode-Based Graphical User Interfaces For Touch Sensitive Input Devices,” filed Jan. 18, 2005; (6) U.S. patent application Ser. No. 11/228,758, “Virtual Input Device Placement On A Touch Screen User Interface,” filed Sep. 16, 2005; (7) U.S. patent application Ser. No. 11/228,700, “Operation Of A Computer With A Touch Screen Interface,” filed Sep. 16, 2005; (8) U.S. patent application Ser. No. 11/228,737, “Activating Virtual Keys Of A Touch-Screen Virtual Keyboard,” filed Sep. 16, 2005; and (9) U.S. patent application Ser. No. 11/367,749, “Multi-Functional Hand-Held Device,” filed Mar. 3, 2006. All of these applications are incorporated by reference herein in their entirety.

(42) Touch screen **112** optionally has a video resolution in excess of 100 dpi. In some embodiments, the touch screen has a video resolution of approximately 160 dpi. The user optionally makes contact with touch screen **112** using any suitable object or appendage, such as a stylus, a finger, and so forth. In some embodiments, the user interface is designed to work primarily with finger-based contacts and gestures, which can be less precise than stylus-based input due to the larger area of contact of a finger on the touch screen. In some embodiments, the device translates the rough finger-based input into a precise pointer/cursor position or command for performing the actions desired by the user.

(43) In some embodiments, in addition to the touch screen, device **100** optionally includes a touchpad for activating or deactivating particular functions. In some embodiments, the touchpad is a touch-sensitive area of the device that, unlike the touch screen, does not display visual output. The touchpad is, optionally, a touch-sensitive surface that is separate from touch screen **112** or an extension of the touch-sensitive surface formed by the touch screen.

(44) Device **100** also includes power system **162** for powering the various components. Power system **162** optionally includes a power management system, one or more power sources (e.g., battery, alternating current (AC)), a recharging system, a power failure detection circuit, a power converter or inverter, a power status indicator (e.g., a light-emitting diode (LED)) and any other components associated with the generation, management and distribution of power in portable devices.

(45) Device **100** optionally also includes one or more optical sensors **164**. FIG. **1**A shows an optical sensor coupled to optical sensor controller **158** in I/O subsystem **106**. Optical sensor **164** optionally includes charge-coupled device (CCD) or complementary metal-oxide semiconductor (CMOS) phototransistors. Optical sensor **164** receives light from the environment, projected through one or more lenses, and converts the light to data representing an image. In conjunction

with imaging module **143** (also called a camera module), optical sensor **164** optionally captures still images or video. In some embodiments, an optical sensor is located on the back of device **100**, opposite touch screen display **112** on the front of the device so that the touch screen display is enabled for use as a viewfinder for still and/or video image acquisition. In some embodiments, an optical sensor is located on the front of the device so that the user's image is, optionally, obtained for video conferencing while the user views the other video conference participants on the touch screen display. In some embodiments, the position of optical sensor **164** can be changed by the user (e.g., by rotating the lens and the sensor in the device housing) so that a single optical sensor **164** is used along with the touch screen display for both video conferencing and still and/or video image acquisition.

(46) Device **100** optionally also includes one or more depth camera sensors **175**. FIG. **1**A shows a depth camera sensor coupled to depth camera controller **169** in I/O subsystem **106**. Depth camera sensor **175** receives data from the environment to create a three dimensional model of an object (e.g., a face) within a scene from a viewpoint (e.g., a depth camera sensor). In some embodiments, in conjunction with imaging module **143** (also called a camera module), depth camera sensor **175** is optionally used to determine a depth map of different portions of an image captured by the imaging module **143**. In some embodiments, a depth camera sensor is located on the front of device **100** so that the user's image with depth information is, optionally, obtained for video conferencing while the user views the other video conference participants on the touch screen display and to capture selfies with depth map data. In some embodiments, the depth camera sensor **175** is located on the back of device, or on the back and the front of the device **100**. In some embodiments, the position of depth camera sensor **175** can be changed by the user (e.g., by rotating the lens and the sensor in the device housing) so that a depth camera sensor **175** is used along with the touch screen display for both video conferencing and still and/or video image acquisition.

(47) Device **100** optionally also includes one or more contact intensity sensors **165**. FIG. **1**A shows a contact intensity sensor coupled to intensity sensor controller **159** in I/O subsystem **106**. Contact intensity sensor **165** optionally includes one or more piezoresistive strain gauges, capacitive force sensors, electric force sensors, piezoelectric force sensors, optical force sensors, capacitive touch-sensitive surfaces, or other intensity sensors (e.g., sensors used to measure the force (or pressure) of a contact on a touch-sensitive surface). Contact intensity sensor **165** receives contact intensity information (e.g., pressure information or a proxy for pressure information) from the environment. In some embodiments, at least one contact intensity sensor is collocated with, or proximate to, a touch-sensitive surface (e.g., touch-sensitive display system **112**). In some embodiments, at least one contact intensity sensor is located on the back of device **100**, opposite touch screen display **112**, which is located on the front of device **100**.

(48) Device **100** optionally also includes one or more proximity sensors **166**. FIG. **1**A shows proximity sensor **166** coupled to peripherals interface **118**. Alternately, proximity sensor **166** is, optionally, coupled to input controller **160** in I/O subsystem **106**. Proximity sensor **166** optionally performs as described in U.S. patent application Ser. No. 11/241,839, "Proximity Detector In Handheld Device"; Ser. No. 11/240,788, "Proximity Detector In Handheld Device"; Ser. No. 11/620,702, "Using Ambient Light Sensor To Augment Proximity Sensor Output"; Ser. No. 11/586,862, "Automated Response To And Sensing Of User Activity In Portable Devices"; and Ser. No. 11/638,251, "Methods And Systems For Automatic Configuration Of Peripherals," which are hereby incorporated by reference in their entirety. In some embodiments, the proximity sensor turns off and disables touch screen **112** when the multifunction device is placed near the user's ear (e.g., when the user is making a phone call).

(49) Device **100** optionally also includes one or more tactile output generators **167**. FIG. **1**A shows a tactile output generator coupled to haptic feedback controller **161** in I/O subsystem **106**. Tactile output generator **167** optionally includes one or more electroacoustic devices such as speakers or other audio components and/or electromechanical devices that convert energy into linear motion

such as a motor, solenoid, electroactive polymer, piezoelectric actuator, electrostatic actuator, or other tactile output generating component (e.g., a component that converts electrical signals into tactile outputs on the device). Contact intensity sensor **165** receives tactile feedback generation instructions from haptic feedback module **133** and generates tactile outputs on device **100** that are capable of being sensed by a user of device **100**. In some embodiments, at least one tactile output generator is collocated with, or proximate to, a touch-sensitive surface (e.g., touch-sensitive display system **112**) and, optionally, generates a tactile output by moving the touch-sensitive surface vertically (e.g., in/out of a surface of device **100**) or laterally (e.g., back and forth in the same plane as a surface of device **100**). In some embodiments, at least one tactile output generator sensor is located on the back of device **100**, opposite touch screen display **112**, which is located on the front of device **100**.
(50) Device **100** optionally also includes one or more accelerometers **168**. FIG. **1**A shows accelerometer **168** coupled to peripherals interface **118**. Alternately, accelerometer **168** is, optionally, coupled to an input controller **160** in I/O subsystem **106**. Accelerometer **168** optionally performs as described in U.S. Patent Publication No. 20050190059, “Acceleration-based Theft Detection System for Portable Electronic Devices,” and U.S. Patent Publication No. 20060017692, “Methods And Apparatuses For Operating A Portable Device Based On An Accelerometer,” both of which are incorporated by reference herein in their entirety. In some embodiments, information is displayed on the touch screen display in a portrait view or a landscape view based on an analysis of data received from the one or more accelerometers. Device **100** optionally includes, in addition to accelerometer(s) **168**, a magnetometer and a GPS (or GLONASS or other global navigation system) receiver for obtaining information concerning the location and orientation (e.g., portrait or landscape) of device **100**.
(51) In some embodiments, the software components stored in memory **102** include operating system **126**, communication module (or set of instructions) **128**, contact/motion module (or set of instructions) **130**, graphics module (or set of instructions) **132**, text input module (or set of instructions) **134**, Global Positioning System (GPS) module (or set of instructions) **135**, and applications (or sets of instructions) **136**. Furthermore, in some embodiments, memory **102** (FIG. **1**A) or **370** (FIG. **3**) stores device/global internal state **157**, as shown in FIGS. **1**A and **3**. Device/global internal state **157** includes one or more of: active application state, indicating which applications, if any, are currently active; display state, indicating what applications, views or other information occupy various regions of touch screen display **112**; sensor state, including information obtained from the device's various sensors and input control devices **116**; and location information concerning the device's location and/or attitude.
(52) Operating system **126** (e.g., Darwin, RTXC, LINUX, UNIX, OS X, iOS, WINDOWS, or an embedded operating system such as VxWorks) includes various software components and/or drivers for controlling and managing general system tasks (e.g., memory management, storage device control, power management, etc.) and facilitates communication between various hardware and software components.
(53) Communication module **128** facilitates communication with other devices over one or more external ports **124** and also includes various software components for handling data received by RF circuitry **108** and/or external port **124**. External port **124** (e.g., Universal Serial Bus (USB), FIREWIRE, etc.) is adapted for coupling directly to other devices or indirectly over a network (e.g., the Internet, wireless LAN, etc.). In some embodiments, the external port is a multi-pin (e.g., 30-pin) connector that is the same as, or similar to and/or compatible with, the 30-pin connector used on iPod® (trademark of Apple Inc.) devices.
(54) Contact/motion module **130** optionally detects contact with touch screen **112** (in conjunction with display controller **156**) and other touch-sensitive devices (e.g., a touchpad or physical click wheel). Contact/motion module **130** includes various software components for performing various operations related to detection of contact, such as determining if contact has occurred (e.g.,

detecting a finger-down event), determining an intensity of the contact (e.g., the force or pressure of the contact or a substitute for the force or pressure of the contact), determining if there is movement of the contact and tracking the movement across the touch-sensitive surface (e.g., detecting one or more finger-dragging events), and determining if the contact has ceased (e.g., detecting a finger-up event or a break in contact). Contact/motion module **130** receives contact data from the touch-sensitive surface. Determining movement of the point of contact, which is represented by a series of contact data, optionally includes determining speed (magnitude), velocity (magnitude and direction), and/or an acceleration (a change in magnitude and/or direction) of the point of contact. These operations are, optionally, applied to single contacts (e.g., one finger contacts) or to multiple simultaneous contacts (e.g., "multitouch"/multiple finger contacts). In some embodiments, contact/motion module **130** and display controller **156** detect contact on a touchpad.
(55) In some embodiments, contact/motion module **130** uses a set of one or more intensity thresholds to determine whether an operation has been performed by a user (e.g., to determine whether a user has "clicked" on an icon). In some embodiments, at least a subset of the intensity thresholds are determined in accordance with software parameters (e.g., the intensity thresholds are not determined by the activation thresholds of particular physical actuators and can be adjusted without changing the physical hardware of device **100**). For example, a mouse "click" threshold of a trackpad or touch screen display can be set to any of a large range of predefined threshold values without changing the trackpad or touch screen display hardware. Additionally, in some implementations, a user of the device is provided with software settings for adjusting one or more of the set of intensity thresholds (e.g., by adjusting individual intensity thresholds and/or by adjusting a plurality of intensity thresholds at once with a system-level click "intensity" parameter).
(56) Contact/motion module **130** optionally detects a gesture input by a user. Different gestures on the touch-sensitive surface have different contact patterns (e.g., different motions, timings, and/or intensities of detected contacts). Thus, a gesture is, optionally, detected by detecting a particular contact pattern. For example, detecting a finger tap gesture includes detecting a finger-down event followed by detecting a finger-up (liftoff) event at the same position (or substantially the same position) as the finger-down event (e.g., at the position of an icon). As another example, detecting a finger swipe gesture on the touch-sensitive surface includes detecting a finger-down event followed by detecting one or more finger-dragging events, and subsequently followed by detecting a finger-up (liftoff) event.
(57) Graphics module **132** includes various known software components for rendering and displaying graphics on touch screen **112** or other display, including components for changing the visual impact (e.g., brightness, transparency, saturation, contrast, or other visual property) of graphics that are displayed. As used herein, the term "graphics" includes any object that can be displayed to a user, including, without limitation, text, web pages, icons (such as user-interface objects including soft keys), digital images, videos, animations, and the like.
(58) In some embodiments, graphics module **132** stores data representing graphics to be used. Each graphic is, optionally, assigned a corresponding code. Graphics module **132** receives, from applications etc., one or more codes specifying graphics to be displayed along with, if necessary, coordinate data and other graphic property data, and then generates screen image data to output to display controller **156**.
(59) Haptic feedback module **133** includes various software components for generating instructions used by tactile output generator(s) **167** to produce tactile outputs at one or more locations on device **100** in response to user interactions with device **100**.
(60) Text input module **134**, which is, optionally, a component of graphics module **132**, provides soft keyboards for entering text in various applications (e.g., contacts **137**, e-mail **140**, IM **141**, browser **147**, and any other application that needs text input).
(61) GPS module **135** determines the location of the device and provides this information for use in various applications (e.g., to telephone **138** for use in location-based dialing; to camera **143** as

picture/video metadata; and to applications that provide location-based services such as weather widgets, local yellow page widgets, and map/navigation widgets).

(62) Applications **136** optionally include the following modules (or sets of instructions), or a subset or superset thereof: Contacts module **137** (sometimes called an address book or contact list); Telephone module **138**; Video conference module **139**; E-mail client module **140**; Instant messaging (IM) module **141**; Workout support module **142**; Camera module **143** for still and/or video images; Image management module **144**; Video player module; Music player module; Browser module **147**; Calendar module **148**; Widget modules **149**, which optionally include one or more of: weather widget **149-1**, stocks widget **149-2**, calculator widget **149-3**, alarm clock widget **149-4**, dictionary widget **149-5**, and other widgets obtained by the user, as well as user-created widgets **149-6**; Widget creator module **150** for making user-created widgets **149-6**; Search module **151**; Video and music player module **152**, which merges video player module and music player module; Notes module **153**; Map module **154**; and/or Online video module **155**.

(63) Examples of other applications **136** that are, optionally, stored in memory **102** include other word processing applications, other image editing applications, drawing applications, presentation applications, JAVA-enabled applications, encryption, digital rights management, voice recognition, and voice replication.

(64) In conjunction with touch screen **112**, display controller **156**, contact/motion module **130**, graphics module **132**, and text input module **134**, contacts module **137** are, optionally, used to manage an address book or contact list (e.g., stored in application internal state **192** of contacts module **137** in memory **102** or memory **370**), including: adding name(s) to the address book; deleting name(s) from the address book; associating telephone number(s), e-mail address(es), physical address(es) or other information with a name; associating an image with a name; categorizing and sorting names; providing telephone numbers or e-mail addresses to initiate and/or facilitate communications by telephone **138**, video conference module **139**, e-mail **140**, or IM **141**; and so forth.

(65) In conjunction with RF circuitry **108**, audio circuitry **110**, speaker **111**, microphone **113**, touch screen **112**, display controller **156**, contact/motion module **130**, graphics module **132**, and text input module **134**, telephone module **138** are optionally, used to enter a sequence of characters corresponding to a telephone number, access one or more telephone numbers in contacts module **137**, modify a telephone number that has been entered, dial a respective telephone number, conduct a conversation, and disconnect or hang up when the conversation is completed. As noted above, the wireless communication optionally uses any of a plurality of communications standards, protocols, and technologies.

(66) In conjunction with RF circuitry **108**, audio circuitry **110**, speaker **111**, microphone **113**, touch screen **112**, display controller **156**, optical sensor **164**, optical sensor controller **158**, contact/motion module **130**, graphics module **132**, text input module **134**, contacts module **137**, and telephone module **138**, video conference module **139** includes executable instructions to initiate, conduct, and terminate a video conference between a user and one or more other participants in accordance with user instructions.

(67) In conjunction with RF circuitry **108**, touch screen **112**, display controller **156**, contact/motion module **130**, graphics module **132**, and text input module **134**, e-mail client module **140** includes executable instructions to create, send, receive, and manage e-mail in response to user instructions. In conjunction with image management module **144**, e-mail client module **140** makes it very easy to create and send e-mails with still or video images taken with camera module **143**.

(68) In conjunction with RF circuitry **108**, touch screen **112**, display controller **156**, contact/motion module **130**, graphics module **132**, and text input module **134**, the instant messaging module **141** includes executable instructions to enter a sequence of characters corresponding to an instant message, to modify previously entered characters, to transmit a respective instant message (for example, using a Short Message Service (SMS) or Multimedia Message Service (MMS) protocol

for telephony-based instant messages or using XMPP, SIMPLE, or IMPS for Internet-based instant messages), to receive instant messages, and to view received instant messages. In some embodiments, transmitted and/or received instant messages optionally include graphics, photos, audio files, video files and/or other attachments as are supported in an MMS and/or an Enhanced Messaging Service (EMS). As used herein, “instant messaging” refers to both telephony-based messages (e.g., messages sent using SMS or MMS) and Internet-based messages (e.g., messages sent using XMPP, SIMPLE, or IMPS).

(69) In conjunction with RF circuitry **108**, touch screen **112**, display controller **156**, contact/motion module **130**, graphics module **132**, text input module **134**, GPS module **135**, map module **154**, and music player module, workout support module **142** includes executable instructions to create workouts (e.g., with time, distance, and/or calorie burning goals); communicate with workout sensors (sports devices); receive workout sensor data; calibrate sensors used to monitor a workout; select and play music for a workout; and display, store, and transmit workout data.

(70) In conjunction with touch screen **112**, display controller **156**, optical sensor(s) **164**, optical sensor controller **158**, contact/motion module **130**, graphics module **132**, and image management module **144**, camera module **143** includes executable instructions to capture still images or video (including a video stream) and store them into memory **102**, modify characteristics of a still image or video, or delete a still image or video from memory **102**.

(71) In conjunction with touch screen **112**, display controller **156**, contact/motion module **130**, graphics module **132**, text input module **134**, and camera module **143**, image management module **144** includes executable instructions to arrange, modify (e.g., edit), or otherwise manipulate, label, delete, present (e.g., in a digital slide show or album), and store still and/or video images.

(72) In conjunction with RF circuitry **108**, touch screen **112**, display controller **156**, contact/motion module **130**, graphics module **132**, and text input module **134**, browser module **147** includes executable instructions to browse the Internet in accordance with user instructions, including searching, linking to, receiving, and displaying web pages or portions thereof, as well as attachments and other files linked to web pages.

(73) In conjunction with RF circuitry **108**, touch screen **112**, display controller **156**, contact/motion module **130**, graphics module **132**, text input module **134**, e-mail client module **140**, and browser module **147**, calendar module **148** includes executable instructions to create, display, modify, and store calendars and data associated with calendars (e.g., calendar entries, to-do lists, etc.) in accordance with user instructions.

(74) In conjunction with RF circuitry **108**, touch screen **112**, display controller **156**, contact/motion module **130**, graphics module **132**, text input module **134**, and browser module **147**, widget modules **149** are mini-applications that are, optionally, downloaded and used by a user (e.g., weather widget **149-1**, stocks widget **149-2**, calculator widget **149-3**, alarm clock widget **149-4**, and dictionary widget **149-5**) or created by the user (e.g., user-created widget **149-6**). In some embodiments, a widget includes an HTML (Hypertext Markup Language) file, a CSS (Cascading Style Sheets) file, and a JavaScript file. In some embodiments, a widget includes an XML (Extensible Markup Language) file and a JavaScript file (e.g., Yahoo! Widgets).

(75) In conjunction with RF circuitry **108**, touch screen **112**, display controller **156**, contact/motion module **130**, graphics module **132**, text input module **134**, and browser module **147**, the widget creator module **150** are, optionally, used by a user to create widgets (e.g., turning a user-specified portion of a web page into a widget).

(76) In conjunction with touch screen **112**, display controller **156**, contact/motion module **130**, graphics module **132**, and text input module **134**, search module **151** includes executable instructions to search for text, music, sound, image, video, and/or other files in memory **102** that match one or more search criteria (e.g., one or more user-specified search terms) in accordance with user instructions.

(77) In conjunction with touch screen **112**, display controller **156**, contact/motion module **130**,

graphics module **132**, audio circuitry **110**, speaker **111**, RF circuitry **108**, and browser module **147**, video and music player module **152** includes executable instructions that allow the user to download and play back recorded music and other sound files stored in one or more file formats, such as MP3 or AAC files, and executable instructions to display, present, or otherwise play back videos (e.g., on touch screen **112** or on an external, connected display via external port **124**). In some embodiments, device **100** optionally includes the functionality of an MP3 player, such as an iPod (trademark of Apple Inc.).

(78) In conjunction with touch screen **112**, display controller **156**, contact/motion module **130**, graphics module **132**, and text input module **134**, notes module **153** includes executable instructions to create and manage notes, to-do lists, and the like in accordance with user instructions.

(79) In conjunction with RF circuitry **108**, touch screen **112**, display controller **156**, contact/motion module **130**, graphics module **132**, text input module **134**, GPS module **135**, and browser module **147**, map module **154** are, optionally, used to receive, display, modify, and store maps and data associated with maps (e.g., driving directions, data on stores and other points of interest at or near a particular location, and other location-based data) in accordance with user instructions.

(80) In conjunction with touch screen **112**, display controller **156**, contact/motion module **130**, graphics module **132**, audio circuitry **110**, speaker **111**, RF circuitry **108**, text input module **134**, e-mail client module **140**, and browser module **147**, online video module **155** includes instructions that allow the user to access, browse, receive (e.g., by streaming and/or download), play back (e.g., on the touch screen or on an external, connected display via external port **124**), send an e-mail with a link to a particular online video, and otherwise manage online videos in one or more file formats, such as H.264. In some embodiments, instant messaging module **141**, rather than e-mail client module **140**, is used to send a link to a particular online video. Additional description of the online video application can be found in U.S. Provisional Patent Application No. 60/936,562, “Portable Multifunction Device, Method, and Graphical User Interface for Playing Online Videos,” filed Jun. 20, 2007, and U.S. patent application Ser. No. 11/968,067, “Portable Multifunction Device, Method, and Graphical User Interface for Playing Online Videos,” filed Dec. 31, 2007, the contents of which are hereby incorporated by reference in their entirety.

(81) Each of the above-identified modules and applications corresponds to a set of executable instructions for performing one or more functions described above and the methods described in this application (e.g., the computer-implemented methods and other information processing methods described herein). These modules (e.g., sets of instructions) need not be implemented as separate software programs (such as computer programs (e.g., including instructions)), procedures, or modules, and thus various subsets of these modules are, optionally, combined or otherwise rearranged in various embodiments. For example, video player module is, optionally, combined with music player module into a single module (e.g., video and music player module **152**, FIG. **1**A). In some embodiments, memory **102** optionally stores a subset of the modules and data structures identified above. Furthermore, memory **102** optionally stores additional modules and data structures not described above.

(82) In some embodiments, device **100** is a device where operation of a predefined set of functions on the device is performed exclusively through a touch screen and/or a touchpad. By using a touch screen and/or a touchpad as the primary input control device for operation of device **100**, the number of physical input control devices (such as push buttons, dials, and the like) on device **100** is, optionally, reduced.

(83) The predefined set of functions that are performed exclusively through a touch screen and/or a touchpad optionally include navigation between user interfaces. In some embodiments, the touchpad, when touched by the user, navigates device **100** to a main, home, or root menu from any user interface that is displayed on device **100**. In such embodiments, a “menu button” is implemented using a touchpad. In some other embodiments, the menu button is a physical push

button or other physical input control device instead of a touchpad.

(84) FIG. **1**B is a block diagram illustrating exemplary components for event handling in accordance with some embodiments. In some embodiments, memory **102** (FIG. **1**A) or **370** (FIG. **3**) includes event sorter **170** (e.g., in operating system **126**) and a respective application **136-1** (e.g., any of the aforementioned applications **137-151**, **155**, **380-390**).

(85) Event sorter **170** receives event information and determines the application **136-1** and application view **191** of application **136-1** to which to deliver the event information. Event sorter **170** includes event monitor **171** and event dispatcher module **174**. In some embodiments, application **136-1** includes application internal state **192**, which indicates the current application view(s) displayed on touch-sensitive display **112** when the application is active or executing. In some embodiments, device/global internal state **157** is used by event sorter **170** to determine which application(s) is (are) currently active, and application internal state **192** is used by event sorter **170** to determine application views **191** to which to deliver event information.

(86) In some embodiments, application internal state **192** includes additional information, such as one or more of: resume information to be used when application **136-1** resumes execution, user interface state information that indicates information being displayed or that is ready for display by application **136-1**, a state queue for enabling the user to go back to a prior state or view of application **136-1**, and a redo/undo queue of previous actions taken by the user.

(87) Event monitor **171** receives event information from peripherals interface **118**. Event information includes information about a sub-event (e.g., a user touch on touch-sensitive display **112**, as part of a multi-touch gesture). Peripherals interface **118** transmits information it receives from I/O subsystem **106** or a sensor, such as proximity sensor **166**, accelerometer(s) **168**, and/or microphone **113** (through audio circuitry **110**). Information that peripherals interface **118** receives from I/O subsystem **106** includes information from touch-sensitive display **112** or a touch-sensitive surface.

(88) In some embodiments, event monitor **171** sends requests to the peripherals interface **118** at predetermined intervals. In response, peripherals interface **118** transmits event information. In other embodiments, peripherals interface **118** transmits event information only when there is a significant event (e.g., receiving an input above a predetermined noise threshold and/or for more than a predetermined duration).

(89) In some embodiments, event sorter **170** also includes a hit view determination module **172** and/or an active event recognizer determination module **173**.

(90) Hit view determination module **172** provides software procedures for determining where a sub-event has taken place within one or more views when touch-sensitive display **112** displays more than one view. Views are made up of controls and other elements that a user can see on the display.

(91) Another aspect of the user interface associated with an application is a set of views, sometimes herein called application views or user interface windows, in which information is displayed and touch-based gestures occur. The application views (of a respective application) in which a touch is detected optionally correspond to programmatic levels within a programmatic or view hierarchy of the application. For example, the lowest level view in which a touch is detected is, optionally, called the hit view, and the set of events that are recognized as proper inputs are, optionally, determined based, at least in part, on the hit view of the initial touch that begins a touch-based gesture.

(92) Hit view determination module **172** receives information related to sub-events of a touch-based gesture. When an application has multiple views organized in a hierarchy, hit view determination module **172** identifies a hit view as the lowest view in the hierarchy which should handle the sub-event. In most circumstances, the hit view is the lowest level view in which an initiating sub-event occurs (e.g., the first sub-event in the sequence of sub-events that form an event or potential event). Once the hit view is identified by the hit view determination module **172**, the

hit view typically receives all sub-events related to the same touch or input source for which it was identified as the hit view.
(93) Active event recognizer determination module **173** determines which view or views within a view hierarchy should receive a particular sequence of sub-events. In some embodiments, active event recognizer determination module **173** determines that only the hit view should receive a particular sequence of sub-events. In other embodiments, active event recognizer determination module **173** determines that all views that include the physical location of a sub-event are actively involved views, and therefore determines that all actively involved views should receive a particular sequence of sub-events. In other embodiments, even if touch sub-events were entirely confined to the area associated with one particular view, views higher in the hierarchy would still remain as actively involved views.
(94) Event dispatcher module **174** dispatches the event information to an event recognizer (e.g., event recognizer **180**). In embodiments including active event recognizer determination module **173**, event dispatcher module **174** delivers the event information to an event recognizer determined by active event recognizer determination module **173**. In some embodiments, event dispatcher module **174** stores in an event queue the event information, which is retrieved by a respective event receiver **182**.
(95) In some embodiments, operating system **126** includes event sorter **170**. Alternatively, application **136-1** includes event sorter **170**. In yet other embodiments, event sorter **170** is a stand-alone module, or a part of another module stored in memory **102**, such as contact/motion module **130**.
(96) In some embodiments, application **136-1** includes a plurality of event handlers **190** and one or more application views **191**, each of which includes instructions for handling touch events that occur within a respective view of the application's user interface. Each application view **191** of the application **136-1** includes one or more event recognizers **180**. Typically, a respective application view **191** includes a plurality of event recognizers **180**. In other embodiments, one or more of event recognizers **180** are part of a separate module, such as a user interface kit or a higher level object from which application **136-1** inherits methods and other properties. In some embodiments, a respective event handler **190** includes one or more of: data updater **176**, object updater **177**, GUI updater **178**, and/or event data **179** received from event sorter **170**. Event handler **190** optionally utilizes or calls data updater **176**, object updater **177**, or GUI updater **178** to update the application internal state **192**. Alternatively, one or more of the application views **191** include one or more respective event handlers **190**. Also, in some embodiments, one or more of data updater **176**, object updater **177**, and GUI updater **178** are included in a respective application view **191**.
(97) A respective event recognizer **180** receives event information (e.g., event data **179**) from event sorter **170** and identifies an event from the event information. Event recognizer **180** includes event receiver **182** and event comparator **184**. In some embodiments, event recognizer **180** also includes at least a subset of: metadata **183**, and event delivery instructions **188** (which optionally include sub-event delivery instructions).
(98) Event receiver **182** receives event information from event sorter **170**. The event information includes information about a sub-event, for example, a touch or a touch movement. Depending on the sub-event, the event information also includes additional information, such as location of the sub-event. When the sub-event concerns motion of a touch, the event information optionally also includes speed and direction of the sub-event. In some embodiments, events include rotation of the device from one orientation to another (e.g., from a portrait orientation to a landscape orientation, or vice versa), and the event information includes corresponding information about the current orientation (also called device attitude) of the device.
(99) Event comparator **184** compares the event information to predefined event or sub-event definitions and, based on the comparison, determines an event or sub-event, or determines or updates the state of an event or sub-event. In some embodiments, event comparator **184** includes

event definitions **186**. Event definitions **186** contain definitions of events (e.g., predefined sequences of sub-events), for example, event 1 (**187-1**), event 2 (**187-2**), and others. In some embodiments, sub-events in an event (**187**) include, for example, touch begin, touch end, touch movement, touch cancellation, and multiple touching. In one example, the definition for event 1 (**187-1**) is a double tap on a displayed object. The double tap, for example, comprises a first touch (touch begin) on the displayed object for a predetermined phase, a first liftoff (touch end) for a predetermined phase, a second touch (touch begin) on the displayed object for a predetermined phase, and a second liftoff (touch end) for a predetermined phase. In another example, the definition for event 2 (**187-2**) is a dragging on a displayed object. The dragging, for example, comprises a touch (or contact) on the displayed object for a predetermined phase, a movement of the touch across touch-sensitive display **112**, and liftoff of the touch (touch end). In some embodiments, the event also includes information for one or more associated event handlers **190**.
(100) In some embodiments, event definition **187** includes a definition of an event for a respective user-interface object. In some embodiments, event comparator **184** performs a hit test to determine which user-interface object is associated with a sub-event. For example, in an application view in which three user-interface objects are displayed on touch-sensitive display **112**, when a touch is detected on touch-sensitive display **112**, event comparator **184** performs a hit test to determine which of the three user-interface objects is associated with the touch (sub-event). If each displayed object is associated with a respective event handler **190**, the event comparator uses the result of the hit test to determine which event handler **190** should be activated. For example, event comparator **184** selects an event handler associated with the sub-event and the object triggering the hit test.
(101) In some embodiments, the definition for a respective event (**187**) also includes delayed actions that delay delivery of the event information until after it has been determined whether the sequence of sub-events does or does not correspond to the event recognizer's event type.
(102) When a respective event recognizer **180** determines that the series of sub-events do not match any of the events in event definitions **186**, the respective event recognizer **180** enters an event impossible, event failed, or event ended state, after which it disregards subsequent sub-events of the touch-based gesture. In this situation, other event recognizers, if any, that remain active for the hit view continue to track and process sub-events of an ongoing touch-based gesture.
(103) In some embodiments, a respective event recognizer **180** includes metadata **183** with configurable properties, flags, and/or lists that indicate how the event delivery system should perform sub-event delivery to actively involved event recognizers. In some embodiments, metadata **183** includes configurable properties, flags, and/or lists that indicate how event recognizers interact, or are enabled to interact, with one another. In some embodiments, metadata **183** includes configurable properties, flags, and/or lists that indicate whether sub-events are delivered to varying levels in the view or programmatic hierarchy.
(104) In some embodiments, a respective event recognizer **180** activates event handler **190** associated with an event when one or more particular sub-events of an event are recognized. In some embodiments, a respective event recognizer **180** delivers event information associated with the event to event handler **190**. Activating an event handler **190** is distinct from sending (and deferred sending) sub-events to a respective hit view. In some embodiments, event recognizer **180** throws a flag associated with the recognized event, and event handler **190** associated with the flag catches the flag and performs a predefined process.
(105) In some embodiments, event delivery instructions **188** include sub-event delivery instructions that deliver event information about a sub-event without activating an event handler. Instead, the sub-event delivery instructions deliver event information to event handlers associated with the series of sub-events or to actively involved views. Event handlers associated with the series of sub-events or with actively involved views receive the event information and perform a predetermined process.
(106) In some embodiments, data updater **176** creates and updates data used in application **136-1**.

For example, data updater **176** updates the telephone number used in contacts module **137**, or stores a video file used in video player module. In some embodiments, object updater **177** creates and updates objects used in application **136-1**. For example, object updater **177** creates a new user-interface object or updates the position of a user-interface object. GUI updater **178** updates the GUI. For example, GUI updater **178** prepares display information and sends it to graphics module **132** for display on a touch-sensitive display.
(107) In some embodiments, event handler(s) **190** includes or has access to data updater **176**, object updater **177**, and GUI updater **178**. In some embodiments, data updater **176**, object updater **177**, and GUI updater **178** are included in a single module of a respective application **136-1** or application view **191**. In other embodiments, they are included in two or more software modules.
(108) It shall be understood that the foregoing discussion regarding event handling of user touches on touch-sensitive displays also applies to other forms of user inputs to operate multifunction devices **100** with input devices, not all of which are initiated on touch screens. For example, mouse movement and mouse button presses, optionally coordinated with single or multiple keyboard presses or holds; contact movements such as taps, drags, scrolls, etc. on touchpads; pen stylus inputs; movement of the device; oral instructions; detected eye movements; biometric inputs; and/or any combination thereof are optionally utilized as inputs corresponding to sub-events which define an event to be recognized.
(109) FIG. **2** illustrates a portable multifunction device **100** having a touch screen **112** in accordance with some embodiments. The touch screen optionally displays one or more graphics within user interface (UI) **200**. In this embodiment, as well as others described below, a user is enabled to select one or more of the graphics by making a gesture on the graphics, for example, with one or more fingers **202** (not drawn to scale in the figure) or one or more styluses **203** (not drawn to scale in the figure). In some embodiments, selection of one or more graphics occurs when the user breaks contact with the one or more graphics. In some embodiments, the gesture optionally includes one or more taps, one or more swipes (from left to right, right to left, upward and/or downward), and/or a rolling of a finger (from right to left, left to right, upward and/or downward) that has made contact with device **100**. In some implementations or circumstances, inadvertent contact with a graphic does not select the graphic. For example, a swipe gesture that sweeps over an application icon optionally does not select the corresponding application when the gesture corresponding to selection is a tap.
(110) Device **100** optionally also include one or more physical buttons, such as “home” or menu button **204**. As described previously, menu button **204** is, optionally, used to navigate to any application **136** in a set of applications that are, optionally, executed on device **100**. Alternatively, in some embodiments, the menu button is implemented as a soft key in a GUI displayed on touch screen **112**.
(111) In some embodiments, device **100** includes touch screen **112**, menu button **204**, push button **206** for powering the device on/off and locking the device, volume adjustment button(s) **208**, subscriber identity module (SIM) card slot **210**, headset jack **212**, and docking/charging external port **124**. Push button **206** is, optionally, used to turn the power on/off on the device by depressing the button and holding the button in the depressed state for a predefined time interval; to lock the device by depressing the button and releasing the button before the predefined time interval has elapsed; and/or to unlock the device or initiate an unlock process. In an alternative embodiment, device **100** also accepts verbal input for activation or deactivation of some functions through microphone **113**. Device **100** also, optionally, includes one or more contact intensity sensors **165** for detecting intensity of contacts on touch screen **112** and/or one or more tactile output generators **167** for generating tactile outputs for a user of device **100**.
(112) FIG. **3** is a block diagram of an exemplary multifunction device with a display and a touch-sensitive surface in accordance with some embodiments. Device **300** need not be portable. In some embodiments, device **300** is a laptop computer, a desktop computer, a tablet computer, a

multimedia player device, a navigation device, an educational device (such as a child's learning toy), a gaming system, or a control device (e.g., a home or industrial controller). Device **300** typically includes one or more processing units (CPUs) **310**, one or more network or other communications interfaces **360**, memory **370**, and one or more communication buses **320** for interconnecting these components. Communication buses **320** optionally include circuitry (sometimes called a chipset) that interconnects and controls communications between system components. Device **300** includes input/output (I/O) interface **330** comprising display **340**, which is typically a touch screen display. I/O interface **330** also optionally includes a keyboard and/or mouse (or other pointing device) **350** and touchpad **355**, tactile output generator **357** for generating tactile outputs on device **300** (e.g., similar to tactile output generator(s) **167** described above with reference to FIG. **1**A), sensors **359** (e.g., optical, acceleration, proximity, touch-sensitive, and/or contact intensity sensors similar to contact intensity sensor(s) **165** described above with reference to FIG. **1**A). Memory **370** includes high-speed random access memory, such as DRAM, SRAM, DDR RAM, or other random access solid state memory devices; and optionally includes non-volatile memory, such as one or more magnetic disk storage devices, optical disk storage devices, flash memory devices, or other non-volatile solid state storage devices. Memory **370** optionally includes one or more storage devices remotely located from CPU(s) **310**. In some embodiments, memory **370** stores programs, modules, and data structures analogous to the programs, modules, and data structures stored in memory **102** of portable multifunction device **100** (FIG. **1**A), or a subset thereof. Furthermore, memory **370** optionally stores additional programs, modules, and data structures not present in memory **102** of portable multifunction device **100**. For example, memory **370** of device **300** optionally stores drawing module **380**, presentation module **382**, word processing module **384**, website creation module **386**, disk authoring module **388**, and/or spreadsheet module **390**, while memory **102** of portable multifunction device **100** (FIG. **1**A) optionally does not store these modules.

(113) Each of the above-identified elements in FIG. **3** is, optionally, stored in one or more of the previously mentioned memory devices. Each of the above-identified modules corresponds to a set of instructions for performing a function described above. The above-identified modules or computer programs (e.g., sets of instructions or including instructions) need not be implemented as separate software programs (such as computer programs (e.g., including instructions)), procedures, or modules, and thus various subsets of these modules are, optionally, combined or otherwise rearranged in various embodiments. In some embodiments, memory **370** optionally stores a subset of the modules and data structures identified above. Furthermore, memory **370** optionally stores additional modules and data structures not described above.

(114) Attention is now directed towards embodiments of user interfaces that are, optionally, implemented on, for example, portable multifunction device **100**.

(115) FIG. **4**A illustrates an exemplary user interface for a menu of applications on portable multifunction device **100** in accordance with some embodiments. Similar user interfaces are, optionally, implemented on device **300**. In some embodiments, user interface **400** includes the following elements, or a subset or superset thereof: Signal strength indicator(s) **402** for wireless communication(s), such as cellular and Wi-Fi signals; Time **404**; Bluetooth indicator **405**; Battery status indicator **406**; Tray **408** with icons for frequently used applications, such as: Icon **416** for telephone module **138**, labeled “Phone,” which optionally includes an indicator **414** of the number of missed calls or voicemail messages; Icon **418** for e-mail client module **140**, labeled “Mail,” which optionally includes an indicator **410** of the number of unread e-mails; Icon **420** for browser module **147**, labeled “Browser;” and Icon **422** for video and music player module **152**, also referred to as iPod (trademark of Apple Inc.) module **152**, labeled “iPod;” and Icons for other applications, such as: Icon **424** for IM module **141**, labeled “Messages;” Icon **426** for calendar module **148**, labeled “Calendar;” Icon **428** for image management module **144**, labeled “Photos;” Icon **430** for camera module **143**, labeled “Camera;” Icon **432** for online video module **155**, labeled “Online

Video;" Icon **434** for stocks widget **149-2**, labeled "Stocks;" Icon **436** for map module **154**, labeled "Maps;" Icon **438** for weather widget **149-1**, labeled "Weather;" Icon **440** for alarm clock widget **149-4**, labeled "Clock;" Icon **442** for workout support module **142**, labeled "Workout Support;" Icon **444** for notes module **153**, labeled "Notes;" and Icon **446** for a settings application or module, labeled "Settings," which provides access to settings for device **100** and its various applications **136**.

(116) It should be noted that the icon labels illustrated in FIG. **4**A are merely exemplary. For example, icon **422** for video and music player module **152** is labeled "Music" or "Music Player." Other labels are, optionally, used for various application icons. In some embodiments, a label for a respective application icon includes a name of an application corresponding to the respective application icon. In some embodiments, a label for a particular application icon is distinct from a name of an application corresponding to the particular application icon.

(117) FIG. **4**B illustrates an exemplary user interface on a device (e.g., device **300**, FIG. **3**) with a touch-sensitive surface **451** (e.g., a tablet or touchpad **355**, FIG. **3**) that is separate from the display **450** (e.g., touch screen display **112**). Device **300** also, optionally, includes one or more contact intensity sensors (e.g., one or more of sensors **359**) for detecting intensity of contacts on touch-sensitive surface **451** and/or one or more tactile output generators **357** for generating tactile outputs for a user of device **300**.

(118) Although some of the examples that follow will be given with reference to inputs on touch screen display **112** (where the touch-sensitive surface and the display are combined), in some embodiments, the device detects inputs on a touch-sensitive surface that is separate from the display, as shown in FIG. **4**B. In some embodiments, the touch-sensitive surface (e.g., **451** in FIG. **4**B) has a primary axis (e.g., **452** in FIG. **4**B) that corresponds to a primary axis (e.g., **453** in FIG. **4**B) on the display (e.g., **450**). In accordance with these embodiments, the device detects contacts (e.g., **460** and **462** in FIG. **4**B) with the touch-sensitive surface **451** at locations that correspond to respective locations on the display (e.g., in FIG. **4**B, **460** corresponds to **468** and **462** corresponds to **470**). In this way, user inputs (e.g., contacts **460** and **462**, and movements thereof) detected by the device on the touch-sensitive surface (e.g., **451** in FIG. **4**B) are used by the device to manipulate the user interface on the display (e.g., **450** in FIG. **4**B) of the multifunction device when the touch-sensitive surface is separate from the display. It should be understood that similar methods are, optionally, used for other user interfaces described herein.

(119) Additionally, while the following examples are given primarily with reference to finger inputs (e.g., finger contacts, finger tap gestures, finger swipe gestures), it should be understood that, in some embodiments, one or more of the finger inputs are replaced with input from another input device (e.g., a mouse-based input or stylus input). For example, a swipe gesture is, optionally, replaced with a mouse click (e.g., instead of a contact) followed by movement of the cursor along the path of the swipe (e.g., instead of movement of the contact). As another example, a tap gesture is, optionally, replaced with a mouse click while the cursor is located over the location of the tap gesture (e.g., instead of detection of the contact followed by ceasing to detect the contact). Similarly, when multiple user inputs are simultaneously detected, it should be understood that multiple computer mice are, optionally, used simultaneously, or a mouse and finger contacts are, optionally, used simultaneously.

(120) FIG. **5**A illustrates exemplary personal electronic device **500**. Device **500** includes body **502**. In some embodiments, device **500** can include some or all of the features described with respect to devices **100** and **300** (e.g., FIGS. **1**A-**4**B). In some embodiments, device **500** has touch-sensitive display screen **504**, hereafter touch screen **504**. Alternatively, or in addition to touch screen **504**, device **500** has a display and a touch-sensitive surface. As with devices **100** and **300**, in some embodiments, touch screen **504** (or the touch-sensitive surface) optionally includes one or more intensity sensors for detecting intensity of contacts (e.g., touches) being applied. The one or more intensity sensors of touch screen **504** (or the touch-sensitive surface) can provide output data that

represents the intensity of touches. The user interface of device **500** can respond to touches based on their intensity, meaning that touches of different intensities can invoke different user interface operations on device **500**.
(121) Exemplary techniques for detecting and processing touch intensity are found, for example, in related applications: International Patent Application Serial No. PCT/US2013/040061, titled "Device, Method, and Graphical User Interface for Displaying User Interface Objects Corresponding to an Application," filed May 8, 2013, published as WIPO Publication No. WO/2013/169849, and International Patent Application Serial No. PCT/US2013/069483, titled "Device, Method, and Graphical User Interface for Transitioning Between Touch Input to Display Output Relationships," filed Nov. 11, 2013, published as WIPO Publication No. WO/2014/105276, each of which is hereby incorporated by reference in their entirety.
(122) In some embodiments, device **500** has one or more input mechanisms **506** and **508**. Input mechanisms **506** and **508**, if included, can be physical. Examples of physical input mechanisms include push buttons and rotatable mechanisms. In some embodiments, device **500** has one or more attachment mechanisms. Such attachment mechanisms, if included, can permit attachment of device **500** with, for example, hats, eyewear, earrings, necklaces, shirts, jackets, bracelets, watch straps, chains, trousers, belts, shoes, purses, backpacks, and so forth. These attachment mechanisms permit device **500** to be worn by a user.
(123) FIG. **5**B depicts exemplary personal electronic device **500**. In some embodiments, device **500** can include some or all of the components described with respect to FIGS. **1**A, **1**B, and **3**. Device **500** has bus **512** that operatively couples I/O section **514** with one or more computer processors **516** and memory **518**. I/O section **514** can be connected to display **504**, which can have touch-sensitive component **522** and, optionally, intensity sensor **524** (e.g., contact intensity sensor). In addition, I/O section **514** can be connected with communication unit **530** for receiving application and operating system data, using Wi-Fi, Bluetooth, near field communication (NFC), cellular, and/or other wireless communication techniques. Device **500** can include input mechanisms **506** and/or **508**. Input mechanism **506** is, optionally, a rotatable input device or a depressible and rotatable input device, for example. Input mechanism **508** is, optionally, a button, in some examples.
(124) Input mechanism **508** is, optionally, a microphone, in some examples. Personal electronic device **500** optionally includes various sensors, such as GPS sensor **532**, accelerometer **534**, directional sensor **540** (e.g., compass), gyroscope **536**, motion sensor **538**, and/or a combination thereof, all of which can be operatively connected to I/O section **514**.
(125) Memory **518** of personal electronic device **500** can include one or more non-transitory computer-readable storage mediums, for storing computer-executable instructions, which, when executed by one or more computer processors **516**, for example, can cause the computer processors to perform the techniques described below, including processes **700** (FIG. **7**) and **900** (FIG. **9**). A computer-readable storage medium can be any medium that can tangibly contain or store computer-executable instructions for use by or in connection with the instruction execution system, apparatus, or device. In some examples, the storage medium is a transitory computer-readable storage medium. In some examples, the storage medium is a non-transitory computer-readable storage medium. The non-transitory computer-readable storage medium can include, but is not limited to, magnetic, optical, and/or semiconductor storages. Examples of such storage include magnetic disks, optical discs based on CD, DVD, or Blu-ray technologies, as well as persistent solid-state memory such as flash, solid-state drives, and the like. Personal electronic device **500** is not limited to the components and configuration of FIG. **5**B, but can include other or additional components in multiple configurations.
(126) As used here, the term "affordance" refers to a user-interactive graphical user interface object that is, optionally, displayed on the display screen of devices **100**, **300**, and/or **500** (FIGS. **1**A, **3**, and **5**A-**5**B). For example, an image (e.g., icon), a button, and text (e.g., hyperlink) each optionally

constitute an affordance.

(127) As used herein, the term “focus selector” refers to an input element that indicates a current part of a user interface with which a user is interacting. In some implementations that include a cursor or other location marker, the cursor acts as a “focus selector” so that when an input (e.g., a press input) is detected on a touch-sensitive surface (e.g., touchpad **355** in FIG. **3** or touch-sensitive surface **451** in FIG. **4**B) while the cursor is over a particular user interface element (e.g., a button, window, slider, or other user interface element), the particular user interface element is adjusted in accordance with the detected input. In some implementations that include a touch screen display (e.g., touch-sensitive display system **112** in FIG. **1**A or touch screen **112** in FIG. **4**A) that enables direct interaction with user interface elements on the touch screen display, a detected contact on the touch screen acts as a “focus selector” so that when an input (e.g., a press input by the contact) is detected on the touch screen display at a location of a particular user interface element (e.g., a button, window, slider, or other user interface element), the particular user interface element is adjusted in accordance with the detected input. In some implementations, focus is moved from one region of a user interface to another region of the user interface without corresponding movement of a cursor or movement of a contact on a touch screen display (e.g., by using a tab key or arrow keys to move focus from one button to another button); in these implementations, the focus selector moves in accordance with movement of focus between different regions of the user interface. Without regard to the specific form taken by the focus selector, the focus selector is generally the user interface element (or contact on a touch screen display) that is controlled by the user so as to communicate the user's intended interaction with the user interface (e.g., by indicating, to the device, the element of the user interface with which the user is intending to interact). For example, the location of a focus selector (e.g., a cursor, a contact, or a selection box) over a respective button while a press input is detected on the touch-sensitive surface (e.g., a touchpad or touch screen) will indicate that the user is intending to activate the respective button (as opposed to other user interface elements shown on a display of the device).

(128) As used in the specification and claims, the term “characteristic intensity” of a contact refers to a characteristic of the contact based on one or more intensities of the contact. In some embodiments, the characteristic intensity is based on multiple intensity samples. The characteristic intensity is, optionally, based on a predefined number of intensity samples, or a set of intensity samples collected during a predetermined time period (e.g., 0.05, 0.1, 0.2, 0.5, 1, 2, 5, 10 seconds) relative to a predefined event (e.g., after detecting the contact, prior to detecting liftoff of the contact, before or after detecting a start of movement of the contact, prior to detecting an end of the contact, before or after detecting an increase in intensity of the contact, and/or before or after detecting a decrease in intensity of the contact). A characteristic intensity of a contact is, optionally, based on one or more of: a maximum value of the intensities of the contact, a mean value of the intensities of the contact, an average value of the intensities of the contact, a top 10 percentile value of the intensities of the contact, a value at the half maximum of the intensities of the contact, a value at the 90 percent maximum of the intensities of the contact, or the like. In some embodiments, the duration of the contact is used in determining the characteristic intensity (e.g., when the characteristic intensity is an average of the intensity of the contact over time). In some embodiments, the characteristic intensity is compared to a set of one or more intensity thresholds to determine whether an operation has been performed by a user. For example, the set of one or more intensity thresholds optionally includes a first intensity threshold and a second intensity threshold. In this example, a contact with a characteristic intensity that does not exceed the first threshold results in a first operation, a contact with a characteristic intensity that exceeds the first intensity threshold and does not exceed the second intensity threshold results in a second operation, and a contact with a characteristic intensity that exceeds the second threshold results in a third operation. In some embodiments, a comparison between the characteristic intensity and one or more thresholds is used to determine whether or not to perform one or more operations (e.g., whether to

perform a respective operation or forgo performing the respective operation), rather than being used to determine whether to perform a first operation or a second operation.
(129) In some embodiments, the device employs intensity hysteresis to avoid accidental inputs sometimes termed “jitter,” where the device defines or selects a hysteresis intensity threshold with a predefined relationship to the press-input intensity threshold (e.g., the hysteresis intensity threshold is X intensity units lower than the press-input intensity threshold or the hysteresis intensity threshold is 75%, 90%, or some reasonable proportion of the press-input intensity threshold). Thus, in some embodiments, the press input includes an increase in intensity of the respective contact above the press-input intensity threshold and a subsequent decrease in intensity of the contact below the hysteresis intensity threshold that corresponds to the press-input intensity threshold, and the respective operation is performed in response to detecting the subsequent decrease in intensity of the respective contact below the hysteresis intensity threshold (e.g., an “up stroke” of the respective press input). Similarly, in some embodiments, the press input is detected only when the device detects an increase in intensity of the contact from an intensity at or below the hysteresis intensity threshold to an intensity at or above the press-input intensity threshold and, optionally, a subsequent decrease in intensity of the contact to an intensity at or below the hysteresis intensity, and the respective operation is performed in response to detecting the press input (e.g., the increase in intensity of the contact or the decrease in intensity of the contact, depending on the circumstances).
(130) For ease of explanation, the descriptions of operations performed in response to a press input associated with a press-input intensity threshold or in response to a gesture including the press input are, optionally, triggered in response to detecting either: an increase in intensity of a contact above the press-input intensity threshold, an increase in intensity of a contact from an intensity below the hysteresis intensity threshold to an intensity above the press-input intensity threshold, a decrease in intensity of the contact below the press-input intensity threshold, and/or a decrease in intensity of the contact below the hysteresis intensity threshold corresponding to the press-input intensity threshold. Additionally, in examples where an operation is described as being performed in response to detecting a decrease in intensity of a contact below the press-input intensity threshold, the operation is, optionally, performed in response to detecting a decrease in intensity of the contact below a hysteresis intensity threshold corresponding to, and lower than, the press-input intensity threshold.
(131) FIG. **5**C illustrates exemplary electronic device **580**. Device **580** includes body **580**A. In some embodiments, device **580** can include some or all of the features described with respect to devices **100**, **300**, and **500** (e.g., FIGS. **1**A-**5**B). In some embodiments, device **580** has one or more speakers **580**B (concealed in body **580**A), one or more microphones **580**C, one or more touch-sensitive surfaces **580**D, and one or more displays **580**E. Alternatively, or in addition to a display and touch-sensitive surface **580**D, the device has a touch-sensitive display (also referred to as a touchscreen). As with devices **100**, **300**, and **500**, in some embodiments, touch-sensitive surface **580**D (or the touch screen) optionally includes one or more intensity sensors for detecting intensity of contacts (e.g., touches) being applied. The one or more intensity sensors of touch-sensitive surface **580**D (or the touchscreen) can provide output data that represents the intensity of touches. The user interface of device **580** can respond to touches based on their intensity, meaning that touches of different intensities can invoke different user interface operations on device **580**. In some embodiments, the one or more displays **580**E are one or more light-emitting diodes (LEDs). For example, a display can be a single LED, an LED cluster (e.g., a red, a green, and a blue LED), a plurality of discrete LEDs, a plurality of discrete LED clusters, or other arrangement of one or more LEDs. For example, the display **580**E can be an array of nine discrete LED clusters arranged in a circular shape (e.g., a ring). In some examples, the one or more displays are comprised of one or more of another type of light-emitting elements.
(132) FIG. **5**D depicts exemplary personal electronic device **580**. In some embodiments, device

580 can include some or all of the components described with respect to FIGS. **1**A, **1**B, **3**, and **5**A-**5**B. Device **580** has bus **592** that operatively couples I/O section **594** with one or more computer processors **596** and memory **598**. I/O section **594** can be connected to display **582**, which can have touch-sensitive component **584** and, optionally, intensity sensor **585** (e.g., contact intensity sensor). In some embodiments, touch-sensitive component **584** is a separate component than display **582**. In addition, I/O section **594** can be connected with communication unit **590** for receiving application and operating system data, using Wi-Fi, Bluetooth, near field communication (NFC), cellular, and/or other wireless communication techniques. Device **580** can include input mechanisms **588**. Input mechanism **588** is, optionally, a button, in some examples. Input mechanism **588** is, optionally, a microphone, in some examples. Input mechanism **588** is, optionally, a plurality of microphones (e.g., a microphone array).

(133) Electronic device **580** includes speaker **586** for outputting audio. Device **580** can include audio circuitry (e.g., in I/O section **594**) that receives audio data, converts the audio data to an electrical signal, and transmits the electrical signal to speaker **586**. Speaker **586** converts the electrical signal to human-audible sound waves. The audio circuitry (e.g., in I/O section **594**) also receives electrical signals converted by a microphone (e.g., input mechanism **588**) from sound waves. The audio circuitry (e.g., in I/O section **594**) converts the electrical signal to audio data. Audio data is, optionally, retrieved from and/or transmitted to memory **598** and/or RF circuitry (e.g., in communication unit **590**) by I/O section **594**.

(134) Memory **598** of personal electronic device **580** can include one or more non-transitory computer-readable storage mediums, for storing computer-executable instructions, which, when executed by one or more computer processors **596**, for example, can cause the computer processors to perform the techniques described below, including processes **700** (FIG. **7**) and **900** (FIG. **9**). A computer-readable storage medium can be any medium that can tangibly contain or store computer-executable instructions for use by or in connection with the instruction execution system, apparatus, or device. In some examples, the storage medium is a transitory computer-readable storage medium. In some examples, the storage medium is a non-transitory computer-readable storage medium. The non-transitory computer-readable storage medium can include, but is not limited to, magnetic, optical, and/or semiconductor storages. Examples of such storage include magnetic disks, optical discs based on CD, DVD, or Blu-ray technologies, as well as persistent solid-state memory such as flash, solid-state drives, and the like. Personal electronic device **580** is not limited to the components and configuration of FIG. **5**D, but can include other or additional components in multiple configurations.

(135) As used herein, an “installed application” refers to a software application that has been downloaded onto an electronic device (e.g., devices **100**, **300**, and/or **500**) and is ready to be launched (e.g., become opened) on the device. In some embodiments, a downloaded application becomes an installed application by way of an installation program that extracts program portions from a downloaded package and integrates the extracted portions with the operating system of the computer system.

(136) As used herein, the terms “open application” or “executing application” refer to a software application with retained state information (e.g., as part of device/global internal state **157** and/or application internal state **192**). An open or executing application is, optionally, any one of the following types of applications: an active application, which is currently displayed on a display screen of the device that the application is being used on; a background application (or background processes), which is not currently displayed, but one or more processes for the application are being processed by one or more processors; and a suspended or hibernated application, which is not running, but has state information that is stored in memory (volatile and non-volatile, respectively) and that can be used to resume execution of the application.

(137) As used herein, the term “closed application” refers to software applications without retained state information (e.g., state information for closed applications is not stored in a memory of the

device). Accordingly, closing an application includes stopping and/or removing application processes for the application and removing state information for the application from the memory of the device. Generally, opening a second application while in a first application does not close the first application. When the second application is displayed and the first application ceases to be displayed, the first application becomes a background application.

(138) Attention is now directed towards embodiments of user interfaces ("UI") and associated processes that are implemented on an electronic device, such as portable multifunction device **100**, device **300**, device **500**, or device **580**.

(139) FIGS. **6**A-**6**V illustrate exemplary user interfaces for managing user requests for specific operations, in accordance with some embodiments. The user interfaces in these figures are used to illustrate the processes described below, including the processes in FIG. **7**.

(140) FIGS. **6**A-**6**F illustrate various user interfaces for configuring user-specific settings relating to personal requests (e.g., requests that require the output of personal data (e.g., media; voicemail; text messages; e-mails; events; to-do items/lists) associated with one or more specific users) within a home network. Each user account associated with a home network has configurable settings that are independent of other users associated with the home network. Devices associated with the home network have configurable settings that are independent of other devices and, in some embodiments, are dependent on the settings of users associated with the home network. In some embodiments, device settings can supersede settings configured for each user account, under certain circumstances as discussed in more detail, below.

(141) In FIG. **6**A, electronic device **600** (e.g., a smart phone) displays, on touchscreen display **602**, time indication **604***a* (showing the current time of 10:00 (e.g., AM)) and home user interface **606**. Device **600** is part of a home network (e.g., a collection of one or more networked devices that are associated with one or more users) called "123 MAIN ST.," which includes a variety of external devices (e.g., lights, speakers, sensors (e.g., smoke detectors; carbon monoxide detectors)) that can be controlled from home user interface **606**. In some embodiments, additional external devices (e.g., door locks, window locks, security cameras, televisions, outlets) are connected to the home network. In particular, device **600** is connected to (e.g., in communication with) three sets of one or more electronic devices for audio output (e.g., smart speakers (e.g., electronic devices **603***a*-**603***d*, as discussed further with respect to FIG. **6**G)) represented by speaker affordances **606***b*, **606***c*, and **606***d* on home user interface **606**. In some embodiments, in response to a set of inputs starting with the selection of speaker affordance **606***b* (e.g., "DINING ROOM SPEAKER"), device **600** displays an external device settings user interface corresponding to the electronic device (e.g., smart speaker) associated with speaker affordance **606***b* (e.g., electronic device **603***b*, "DINING ROOM SPEAKER," of FIG. **6**G). In some embodiments, the external device settings user interface includes configurable settings for enabling or disabling voice recognition (e.g., identifying a user after receiving a voice input (e.g., speech input)). In some embodiments, if a speaker has voice recognition disabled, the speaker's voice cannot be used to make personal requests, which are requests that provide information that is unique to the speaker. Personal requests which will be discussed with greater detail below. In some embodiments, similar external device settings user interfaces are displayed in response to selection of speaker affordances **606***c* and **606***d*. In some embodiments, device **600** includes one or more features of devices **100**, **300**, and/or **500**. In some embodiments, devices **603***a*-**603***d* include one or more features of devices **100**, **300**, **500**, and/or **580**.

(142) Home user interface **606** of FIG. **6**A further includes settings affordance **606***a*. Device **600** detects tap input **608** corresponding to selection of settings affordance **606***a*. In response to detecting tap input **608**, device **600** displays settings user interface **610**, as shown in FIG. **6**B. In some embodiments, in response to detecting a tap input corresponding to selection of settings affordance **606***a*, device **600** displays a menu for navigating to either settings user interface **610** or, if device **600** is authorized to access additional homes (e.g., smart home networks), additional

home user interfaces similar to home user interface **606**. In some embodiments, device **600** receives a set of tap inputs that includes tap **608** and, in response to receiving the set of tap inputs, displays settings user interface **610**.

(143) In FIG. **6**B, device **600** displays settings user interface **610** on touchscreen display **602**. Settings user interface **610** includes a variety of configurable settings relating to the home network for "123 MAIN ST." In particular, settings user interface **610** includes user affordance **610***a*, representing that user "JOHN APPLESEED" is a member of the home and user affordance **610***b*, representing that user "JANE APPLESEED" is also a member of the home. In this example, device **600** and home network "123 MAIN ST." are associated with John (e.g., John is the primary user of home network "123 MAIN ST." John may have initially created home network "123 MAIN ST."). In some embodiments, in response to detecting a set of tap inputs starting on the "INVITE PEOPLE" affordance, device **600** sends invitations to contacts (e.g., family members, friends) to become members of home network "123 MAIN ST." In FIG. **6**B, Jane has previously accepted an invitation to become a member of home network "123 MAIN ST.", as indicated by Jane's user affordance **612***b* shown on settings user interface **610**.

(144) Device **600** detects tap input **612** corresponding to selection of John's user affordance **610***a*. In response to receiving tap input **612**, device **600** displays personal settings user interface **614** on touchscreen display **602**, as shown in FIG. **6**C. Personal settings user interface **614** includes various user settings (e.g., digital assistant settings; media settings; remote access settings; notification settings; and/or home security settings) corresponding to John's user account, including voice recognition setting **614***a*, shown toggled to the "ON" position, and personal request setting **614***b*, shown as "OFF." Voice recognition setting **614***a* and personal request setting **614***b* are settings that configure how an electronic device (e.g., a device associated with John's user account, such as speaker **603***a*) processes user interactions within the home network for the displayed user account (e.g., John's user account). In some embodiments, in response to a tap input corresponding to Jane's user affordance **610***b*, device **600** displays a personal settings user interface corresponding to Jane's user account, similar to personal settings user interface **614** corresponding to John's user account in FIG. **6**C.

(145) In FIG. **6**C, device **600** detects tap input **616** corresponding to selection of personal request setting **614***b*. In response to receiving tap input **616**, device **600** displays personal request user interface **618** associated with John's user account on touchscreen display **602**, as shown in FIG. **6**D. Personal request user interface **618** includes speaker affordance **618***a* (e.g., "BEDROOM SPEAKER"), speaker affordance **618***b* (e.g., "DINING ROOM SPEAKER"), and speaker affordance **618***c* (e.g., "LIVING ROOM SPEAKERS"). Each of the speaker affordances correspond to a set of one or more electronic devices (e.g., smart speakers) connected to home network "123 MAIN ST." As shown in FIG. **6**D, each of the speaker affordances **618***a*-**618***c* include toggles that are in the "OFF" position, indicating that the electronic devices (e.g., smart speakers) associated with speaker affordances **618***a*-**618***c* are not currently configured to perform (e.g., process) personal requests for John (e.g., when the electronic device recognizes that a speech input was from John (e.g., via voice recognition)). In some embodiments, the electronic devices (e.g., smart speakers) associated with speaker affordances **618***a*-**618***c* shown on personal request user interface **618** have voice recognition enabled and, therefore, can process personal requests. In some embodiments, electronic devices (e.g., smart speakers) that have voice recognition disabled do not have a corresponding speaker affordance shown on personal request user interface **618**. In some embodiments, for an electronic device (e.g., smart speaker) that has voice recognition disabled, personal request user interface **618** includes a corresponding speaker affordance that is not configurable (e.g., disabled; cannot be modified (e.g., is shown as grayed out; is shown as blurred)).

(146) In FIG. **6**D, device **600** receives tap inputs **620***a* and **620***b* corresponding to the selection of the toggles in speaker affordances **618***a* and **618***b*, respectively. As shown in FIG. **6**E, in response

to receiving tap inputs **620***a* and **620***b*, device **600** continues displaying personal request user interface **618**, now with speaker affordances **618***a* and **618***b* including toggles that are in the "ON" position. Speaker affordances **618***a* and **618***b* with toggles that are in the "ON" position indicate that the electronic devices (e.g., smart speakers) associated with speaker affordances **618***a* and **618***b* are able to perform (e.g., process) personal requests for John (e.g., when the electronic device recognizes that a speech input was from John (e.g., via voice recognition)).

(147) In FIG. **6**E, in addition to including speaker affordances **618***a*-**618***c*, personal request user interface **618** includes activity notification affordance **618***d*. Activity notification affordance **618***d* is included on personal request user interface **618** when personal requests are enabled for one or more electronic devices (e.g., smart speakers) for the given user, as currently indicated by speaker affordances **618***a* and **618***b* having toggles in the "ON" position. Activity notification affordance **618***d* includes a toggle that is in the "ON" position, indicating that device **600** will display a notification when a personal request has been performed by the electronic devices (e.g., smart speaker) associated with speaker affordances **618***a* and **618***b*. In some embodiments, activity notifications can be enabled per electronic device (e.g., device **600** displays a notification for a personal request completed by the bedroom speaker, but does not display a notification for a personal request completed by the dining room speaker). In some embodiments, activity notifications can provide users with better feedback about requests that are being made for their personal information, which can also improve security and privacy (e.g., by notifying users of unauthorized requests and disclosure of personal information). In FIG. **6**E, device **600** detects tap input **622** corresponding to selection of the toggle in activity notification affordance **618***d*.

(148) As shown in FIG. **6**F, in response to receiving tap input **622**, device **600** displays, on touchscreen display **602**, personal request user interface **618** with the toggle of activity notification affordance **618***d* in the "OFF" position. Activity notification affordance **618***d* showing the toggle in the "OFF" position indicates that device **600** will not display a notification (e.g., notifications are disabled) when a personal request has been performed by the electronic devices (e.g., smart speaker) associated with speaker affordances **618***a* and **618***b*. In some embodiments, when personal requests are first enabled for an electronic device (e.g., smart speaker), activity notification affordance **618***d* is shown with the toggle in the "OFF" position, as in FIG. **6**F, rather than in the "ON" position shown in FIG. **6**E. In some embodiments, selection of the toggle in the "OFF" position in activity notification affordance **618***d* re-enables notifications.

(149) In some embodiments, an electronic device (e.g., smart phone) belonging to Jane Appleseed displays personal request settings associated with Jane's user account analogous to those configurable in personal request user interface **618** associated with John's user account. In some embodiments, the personal request settings configured for a first user account are independent of the personal request settings configured for a second user account (e.g., John's personal request settings are unaffected by Jane's personal request settings), and therefore, multiple user accounts can have different configurations for their respective personal request settings.

(150) Turning now to FIGS. **6**G-**6**V, user interactions with electronic devices (e.g., smart phone, smart speakers) will be described using the various example configurations of personal requests settings described with respect to FIGS. **6**D-**6**F. Starting with FIGS. **6**G-**6**P, settings illustration **605***a* corresponds to the example personal requests settings configuration shown in FIG. **6**E. For FIGS. **6**S-**6**T, settings illustration **605***b* corresponds to the example personal requests settings configuration shown in FIG. **6**F. For FIGS. **6**U-**6**V, settings illustration **605***c* corresponds to the example personal requests settings configuration shown in FIG. **6**D. Most of FIGS. **6**G-**6**V include a schematic diagram of a home (e.g., "123 MAIN ST.") and representations of devices relative to physical properties of the home (e.g., a location of a device relative to a wall or room of the home). The depicted devices are connected to home network "123 MAIN ST." and are in communication with each other via wireless connections. For example, home schematic **601** in FIG. **6**G includes a user indication (e.g., a symbol of a person (e.g., John)) at a location within the home (e.g., in the

bedroom) for interacting with devices (e.g., device **603***a*) within a particular room. The figures and schematic diagram are not necessarily to scale and are provided for exemplary purposes only as a visual aid for the description.

(151) FIG. **6**G illustrates device **600** with touchscreen display **602** off (e.g., device **600** is in a standby mode; device **600** is not displaying a user interface), along with home schematic **601** and settings illustration **605***a*, which corresponds to the example personal requests settings configuration shown in FIG. **6**E. Home schematic **601** is a representation of the home associated with home network "123 MAIN ST." and includes device **600** (e.g., a smart phone) on a counter in the dining room, device **603***a* (e.g., a smart speaker) and device **607** (e.g., a tablet, a smart phone) in the bedroom, device **603***b* (e.g., a smart speaker) in the dining room, and devices **603***c* and **603***d* (e.g., smart speakers) in the living room, all of which are connected to home network "123 MAIN ST." In some embodiments, devices **603***a*-**603***d* include one or more input devices, such as a microphone, and one or more output devices, such as speakers (e.g., electroacoustic speakers). In some embodiments, devices **603***a*-**603***d* include a digital assistant for receiving and processing user commands (e.g., natural language speech input; input via device **600**). Devices **603***c* and **603***d* are smart speakers that are paired in a stereo configuration. In some embodiments, speakers in a stereo configuration share the same settings (e.g., both speakers have voice recognition enabled; both speakers have personal requests disabled) and operations (e.g., both speakers are off; both speakers are playing media). Since devices **603***c* and **603***d* are paired in a stereo configuration, speaker affordance **606***c* on home user interface **606** of FIG. **6**A provides controls for both devices and speaker affordance **618***c* on personal request user interface **618** of FIGS. **6**D-**6**F that configures personal request settings for both devices.

(152) In FIG. **6**G, as shown in home schematic **601**, device **603***a* receives speech input **626***a*, "Assistant, read my last message," from John in the bedroom. Speech input **626***a* is a personal request requiring device **603***a* to access John's latest text message. In some embodiments, after receiving speech input **626***a*, device **603***a* determines the type of request for speech input **626***a* based on a set of criteria that includes a criterion that is met when the request is for user-specific data (e.g., a personal request (e.g., to access a text message; to access a calendar appointment; to access a reminder; to access a note; to access an e-mail; to access a call log). As shown in FIG. **6**H, in response to receiving speech input **626***a*, device **603***a* provides natural-language response **626***b*, "Jane says, 'Have a nice dentist appointment'," because device **603***a* is enabled to process personal requests for John's user account, as shown in settings illustration **605***a*. In some embodiments, prior to providing natural-language response **626***b* and after determining speech input **626***a* is a personal request, device **603***a* performs voice recognition operations to identify speech input **626***a* as matching a voice profile associated with John's user account. In some embodiments, if the voice recognition operations are unsuccessful in matching the speech input with a voice profile associated with a user, device **603***a* forgoes providing natural-language response **626***b* and, in some embodiments, provides a selected natural-language response other than the requested operation. In some embodiments, the natural-language response details that the request cannot be completed (e.g., "Sorry, I can't complete that request.") or includes a query for device **603***a* to identify the user (e.g., "Who's last message?").

(153) In some embodiments, after identifying that the speech input matches the voice profile associated with John's user account, device **603***a* performs an authentication operation to confirm that device **600** (e.g., John's phone) is within proximity of device **603***a* (e.g., by confirming device **600** is connected to the same home network as device **603***a*). In some embodiments, if the authentication operation performed by device **603***a* is unsuccessful (e.g., device **600** is not within proximity), device **603***a* forgoes providing natural-language response **626***b* and, in some embodiments, provides a natural-language response prompting correction of the error (e.g., "Before I can perform your request, please connect your phone to the home network"). In some embodiments, voice recognition operations and/or authentication operations are performed by

device **603***a* on a per-region basis (e.g., location (e.g., within a country)) of the home network. In some embodiments, device **603***a* will not perform voice recognition operations if the home network is configured in a region that does not offer voice recognition capabilities. In some embodiments, devices **603***b*, **603***c*, and **603***d* perform voice recognition operations and/or authentication operations after receiving a speech input, similar to those detailed with respect to device **603***a*.

(154) In FIG. **6**H, upon device **603***a* providing natural-language response **626***b*, device **600** displays lock screen user interface **624** containing bedroom speaker notification **626***c* on touchscreen display **602**. Lock screen user interface **624** includes time indication **604***b* (showing the current time of 10:10 (e.g., AM)), message notification **626***d* issued "30 MINS AGO," and bedroom speaker notification **626***c* issued "NOW." Message notification **626***d* corresponds to a text message from Jane Appleseed and is the last message received by device **600**, as detailed by device **603***a* in natural-language response **626***b* in response to speech input **626***a*, "Assistant, read my last message.," of FIG. **6**G. Since device **600** is in a locked state, as indicated by the locked padlock icon on lock screen user interface **624**, message notification **626***d* is also shown in a locked state (e.g., shows the sender (e.g., Jane Appleseed) without including the message content (e.g., "Have a nice dentist appointment.").

(155) In FIG. **6**H, device **600** displays bedroom speaker notification **626***c* on lock screen user interface **624** because notifications for personal request activity is "ON" for John's user account, as shown in settings illustration **605***a* and activity notification affordance **618***d* of FIG. **6**E. Bedroom speaker notification **626***c* is an indication that device **603***a* (e.g., bedroom speaker) provided a response to speech input **626***a*, which was determined to be a personal request. Bedroom speaker notification **626***c* includes a recency indication (e.g., "NOW"), which indicates when the personal request was processed by device **603***a* and ages as time passes, and text detailing the application that was accessed to complete the personal request (e.g., "MESSAGES").

(156) Home schematic **601** includes electronic device **607** (e.g., a tablet, a smart phone), which is John's tablet that is connected to the home network. In some embodiments, device **607** can be configured to display the same notifications as device **600** and, therefore, displays at least bedroom speaker notification **626***c* and message notification **626***d*. However, in some embodiments, despite notification configurations for device **607**, device **607** does not display bedroom speaker notification **626***c* because activity notifications are enabled for a primary device (e.g., device **600**; not device **607**) that was used to configure the user profile (e.g., John's user profile) and personal request settings. In some embodiments, personal requests and activity notifications are configured at a primary device (e.g., device **600** (e.g., smart phone)) for a specific user and cannot be configured at a secondary device (e.g., device **607** (e.g., tablet)).

(157) Turning now to FIG. **6**I, device **600** continues displaying lock screen user interface **624** on touchscreen display **602**. In home schematic **601**, device **603***a* receives speech input **628***a*, "Assistant, when's my dentist appointment?" from John in the bedroom. Speech input **628***a* is a personal request requiring device **603***a* to access John's calendar. In FIG. **6**J, in response to receiving speech input **628***a*, device **603***a* provides natural-language response **628***b*, "Your dentist appointment is at 12 PM," because device **603***a* is enabled to process personal requests for John's user account, as shown in settings illustration **605***a*. In some embodiments, prior to providing natural-language response **628***b* and after determining speech input **628***a* is a personal request, device **603***a* performs voice recognition operations to identify speech input **628***a* as matching a voice profile associated with John's user account. In some embodiments, if the voice recognition operations are unsuccessful in matching the speech input with a voice profile associated with a user, device **603***a* forgoes providing natural-language response **628***b* and, in some embodiments, provides a natural-language response. In some embodiments, the natural-language response details that the request cannot be completed (e.g., "Sorry, I can't complete that request.") or includes a query for device **603***a* to identify the user (e.g., "Who's last message?").

(158) In FIG. **6**J, lock screen user interface **624** is updated to include bedroom speaker notification

628*c*, along with bedroom speaker notification **626***c* and message notification **626***d*. Bedroom speaker notification **628***c* includes a recency indication (e.g., “NOW”) to indicate when the personal request was processed by device **603***a* and text detailing what application was accessed to complete the personal request (e.g., “CALENDAR”). Bedroom speaker notification **628***c* and bedroom speaker notification **626***c* are displayed as a group (e.g., a stack; a cluster) of notifications with bedroom speaker notification **628***c* overlaid on bedroom speaker notification **626***c*. Device **600** displays bedroom speaker notification **628***c* overlaid on bedroom speaker notification **626***c* to indicate that bedroom speaker notification **628***c* corresponds to the most recent personal request completed by device **603***a* (e.g., bedroom speaker). In some embodiments, additional bedroom speaker notifications are included in the notification stack for additional personal requests completed by device **603***a* (e.g., bedroom speaker).

(159) Turning now to FIG. **6**K, device **600** continues to display lock screen user interface **624** on touchscreen display **602** with bedroom speaker notifications **628***c* and **626***c* and message notification **626***d*. One minute has passed, as indicated by time indication **604***c* (showing the current time of 10:11 (e.g., AM)) and the updated recency indications of bedroom speaker notification **628***c* (e.g., “1 MIN AGO”) and message notification **626***d* (e.g., “31 MINS AGO”). As shown in home schematic **601**, device **603***a* received speech input **630***a*, “Assistant, what's the weather like?”

(160) In FIG. **6**L, device **603***a* provides natural-language response **630***b*, “The high today will be 74 degrees,” in response to receiving speech input **630***a*. Speech input **630***a* is a non-personal request requiring device **603***a* to provide a weather forecast. In some embodiments, in response to receiving speech input **630***a* and prior to providing natural-language response **630***b*, device **603***a* determines the type of request for speech input **630***a* is not a personal request. In some embodiments, if the request is not a personal request, device **603***a* does not perform voice recognition operations and/or authentication operations.

(161) In FIG. **6**L, device **600** continues to display lock screen user interface **624** on touchscreen display **602** without changes. Lock screen user interface **624** includes bedroom speaker notifications **628***c* and **626***c* and message notification **626***d* and does not include a new bedroom speaker notification corresponding to natural-language response **630***b*. For completion of requests that are determined to be non-personal requests (e.g., speech input **630***a*, “Assistant, what's the weather like?”) by device **603***a*, a bedroom speaker notification is not issued for display on device **600**.

(162) Turning now to FIG. **6**M, device **600** maintains display of lock screen user interface **624** on touchscreen display **602** without changes to notifications **628***c*, **626***c*, and **626***d*. In home schematic **601**, the user indication for John is shown in the living room. Paired devices **603***c* and **603***d* receive speech input **632***a*, “Assistant, what's on my grocery list?” from John in the living room. Speech input **632***a* is a personal request requiring paired devices **603***c* and **603***d* to access John's reminders. In some embodiments, in response to receiving speech input **632***a*, paired devices **603***c* and **603***d* determine the type of request for speech input **632***a* is a personal request. In some embodiments, after determining the type of request for speech input **632***a* is a personal request, paired devices **603***c* and **603***d* perform voice recognition operations to identify speech input **632***a* as matching a voice profile associated with John's user account.

(163) In FIGS. **6**M and **6**N, according to settings illustration **605***a*, the living room speakers (e.g., paired devices **603***c* and **603***d*) are not currently configured to process personal requests for John's user account. Therefore, in response to receiving speech input **632***a*, paired devices **603***c* and **603***d* provide natural-language response **632***b*, “Before I can read your reminders, update your settings in the Home App,” via device **603***c* in FIG. **6**N. In some embodiments, paired devices **603***c* and **603***d* each provide audio output for natural-language response **632***b*. In some embodiments, devices that are unable to process personal requests for a particular user account provide a natural-language response to update the settings for personal requests for the particular user account. In some

embodiments, since user settings are independently configured, devices that are unable to process personal request for a first user (e.g., personal requests for John's user account is disabled on paired devices **603***c* and **603***d*) can process personal requests for a second user (e.g., personal requests for Jane's user account can be enabled on paired devices **603***c* and **603***d*). In FIGS. **6**M and **6**N, device **600** does not display a notification corresponding to request **632***a* because the requested operation was not performed. In some embodiments, a notification is displayed on device **600** in response to request **632***a*, even though the requested operation was not performed (e.g., to provide John with feedback that a personal request was made).

(164) Turning now to FIG. **6**O, device **600** maintains display of lock screen user interface **624** on touchscreen display **602** without changes to notifications **628***c*, **626***c*, and **626***d*. In home schematic **601**, the user indication for John is shown in the dining room. Device **603***b* receives speech input **634***a*, "Assistant, what's on my grocery list?" from John in the dining room. Speech input **634***a* is a personal request requiring device **603***b* to access John's reminders, similar to speech input **632***a* of FIG. **6**M. In FIG. **6**P, in response to receiving speech input **634***a*, device **603***b* provides natural-language response **634***b*, "Your grocery list has milk, eggs, and sugar," because device **603***b* is enabled to process personal requests for John's user account, as shown in settings illustration **605***a*. In some embodiments, prior to providing natural-language response **634***b* and after determining speech input **634***a* is a personal request, device **603***b* performs voice recognition operations to identify speech input **634***a* as matching a voice profile associated with John's user account. In some embodiments, if the voice recognition operations are unsuccessful in matching the speech input with a voice profile associated with a user, device **603***b* forgoes providing natural-language response **634***b* and, in some embodiments, provides a natural-language response. In some embodiments, the natural-language response details that the request cannot be completed (e.g., "Sorry, I can't complete that request.") or includes a query for device **603***b* to identify the user (e.g., "Who's last message?").

(165) In FIG. **6**P, device **600** displays, on touchscreen display **602**, lock screen user interface **624**. Lock screen user interface **624** includes new dining room speaker notification **634***c*, along with bedroom speaker notifications **628***c* and **626***c* and message notification **626***d*. Dining room speaker notification **634***c* is shown on lock screen user interface **624** because notifications for personal request activity is "ON" for John's user account, as shown in settings illustration **605***a* and activity notification affordance **618***d* of FIG. **6**E. Dining room speaker notification **634***c* is an indication that device **603***b* (e.g., dining room speaker) provided a response to speech input **634***a*, which was determined to be a personal request, by accessing user data stored in the reminders application (e.g. "REMINDERS"). Dining room speaker notification **634***c* also includes a recency indication (e.g., "NOW"). Unlike bedroom speaker notification **628***c* that is overlaid on bedroom speaker notification **626***c*, dining room speaker notification **634***c* is shown separately from the bedroom speaker notification stack. In some embodiments, additional personal requests performed at device **603***b* (e.g., dining room speaker) causes device **600** to display additional dining room speaker notifications in a stacked arrangement. Notifications for personal requests are grouped (e.g., stacked; in a compressed arrangement) based on the device (e.g., device **603***a*; device **603***b*) used to complete the personal request, with the most recent notification shown overlaid on prior notification. In some embodiments, all notifications for personal requests (e.g., from all devices) are arranged in a stack (e.g., grouped; compressed). In some embodiments, notifications for personal requests are arranged in a stack (e.g., grouped; compressed) by the application (e.g., messages, calls history, calendar, reminders) accessed to complete the personal request.

(166) FIGS. **6**Q-**6**R illustrate notification behavior in response to user input at device **600**. In FIG. **6**Q, device **600** displays, on touchscreen display **602**, lock screen user interface **624** with dining room speaker notification **634***c*, bedroom speaker notifications **628***c* and **626***c*, and message notification **626***d*. Lock screen user interface **624** further includes time indication **604***d* (showing the current time of 10:12 (e.g., AM)) and, for each notification, the recency indications progress by

1 minute. Device **600** is in an unlocked (e.g., authenticated) state, as indicated by the unlocked padlock icon on lock screen user interface **624**. When device **600** is in an unlocked state, message notification **626***d* shows the text message from Jane Appleseed, “Have a nice dentist appointment.” Device **600** detects input **636** (e.g., a tap input; a long-press input) corresponding to selection of dining room speaker notification **634***c*.

(167) As shown in FIG. **6**R, in response to input **636** at dining room speaker notification **634***c*, device **600** displays, on touchscreen display **602**, dining room speaker notification **634***c* and personal request settings affordance **638** overlaid on lock screen user interface **624**. In some embodiments, input **636** is a tap input corresponding to selection of dining room speaker notification **634***c*. In some embodiments, input **636** is a long-press input (e.g., contact with touchscreen display **602** for longer than a threshold amount of time) corresponding to selection of dining room speaker notification **634***c*. In some embodiments, device **600** displays lock screen user interface **624** with a deemphasizing visual treatment (e.g., blur effect; dim effect). In some embodiments, in response to detecting selection of personal request settings affordance **638**, device **600** displays personal request user interface **618**, as shown in FIGS. **6**D-**6**F. In some embodiments, in addition to displaying personal request settings affordance **638**, device **600** displays other selectable affordances that, when selected, cause device **600** to display various setting user interfaces (e.g., dining room speaker settings; home settings).

(168) Turning back to FIG. **6**Q, in some embodiments, in response to detecting a tap input corresponding to selection of the stacked bedroom speaker notifications **628***c* and **626***c*, device **600** displays bedroom speaker notifications **628***c* and **626***c* in an expanded arrangement such that bedroom speaker notification **628***c* and bedroom speaker notification **626***c* are not overlapping and each independently selectable. In such embodiments, in response to selection of either bedroom speaker notification **628***c* or bedroom speaker notification **626***c*, device **600** displays a user interface emphasizing the selected bedroom speaker notification with personal request settings affordance **638**, analogous to that of FIG. **6**R. In some embodiments, in response to in response to detecting a tap input corresponding to selection of the stacked bedroom speaker notifications **628***c* and **626***c*, device **600** displays a user interface emphasizing bedroom speaker notification **628***c* with personal request settings affordance **638**, analogous to that of FIG. **6**R. In some embodiments, in response to detecting a long-press (e.g., tap and hold) input corresponding to selection of the stacked bedroom speaker notifications **628***c* and **626***c*, device **600** displays a user interface emphasizing bedroom speaker notification **628***c* with personal request settings affordance **638**, analogous to that of FIG. **6**R.

(169) FIGS. **6**S-**6**T illustrate a scenario when personal request activity notifications are turned “OFF” for John's user account, as shown in settings illustration **605***b*. Settings illustration **605***b* corresponds to the configuration described with respect to FIG. **6**F. Except for the change in settings from **605***a* to **605***b*, FIG. **6**S is the same as FIG. **6**G. In FIG. **6**S, the touchscreen display **602** of device **600** is off (e.g., device **600** is in a standby mode; device **600** is not displaying a user interface). In home schematic **601**, the user indication for John is shown in the bedroom. Device **603***a* receives speech input **626***a*, “Assistant, read my last message,” from John in the bedroom. Speech input **626***a* is a personal request requiring device **603***a* to access John's latest text message. In some embodiments, after receiving speech input **626***a*, device **603***a* determines the type of request for speech input **626***a* is a personal request. As shown in FIG. **6**T, in response to receiving speech input **626***a*, device **603***a* provides natural-language response **626***b*, “Jane says, ‘Have a nice dentist appointment’,” because device **603***a* is enabled to process personal requests for John's user account, as shown in settings **605***b*. In some embodiments, prior to providing natural-language response **626***b* and after determining speech input **626***a* is a personal request, device **603***a* performs voice recognition operations and/or authorization operations, as discussed in greater detail with respect to FIGS. **6**G-**6**H.

(170) In FIG. **6**T, since personal request activity notifications are disabled (e.g., turned off) for

John's user account, device **600** remains in standby mode with touchscreen display **602** off. Unlike FIG. **6**H (in which device **600** displayed bedroom speaker notification **626***c*), device **600** forgoes display of a bedroom speaker notification indicating that a personal request was completed (e.g., processed) by device **603***a* (e.g., bedroom speaker).

(171) FIGS. **6**U-**6**V illustrate a scenario when personal requests have not yet been configured for a particular user (e.g., after setting up a new external device (e.g., smart speaker); after updating software)). As shown on FIG. **6**V, device **600** displays lock screen user interface **624** with time indication **604***e* (showing the current time of 8:00 (e.g., AM)), which is prior to the time indications throughout FIGS. **6**A-**6**R and, therefore, occurs before personal requests are configured on device **600** for John's user account. In FIG. **6**U, settings **605***c* indicate personal requests are "OFF" (e.g., disabled; not yet setup) for all sets of one or more speakers in home schematic **601**, and personal request activity notifications are "OFF." In some embodiments, the devices (e.g., **603***a*-**603***d*) must be configured to perform specific operations (e.g., voice identification operations; authentication operations) prior to being able to process personal requests from users within a home network.

(172) In FIG. **6**U, touchscreen display **602** of device **600** is off (e.g., device **600** is in a standby mode; device **600** is not displaying a user interface). In home schematic **601**, the user indication for John is shown in the dining room. Device **603***b* receives speech input **640***a*, "Assistant, what does my day look like?" from John in the dining room. Speech input **640***a* is a personal request requiring device **603***b* to access John's user data in one or more applications (e.g., calendar, reminders).

(173) In FIG. **6**V, in response to receiving speech input **640***a*, device **603***b* provides natural-language response **640***b*, "Before I can help you with that, you'll need to set up personal requests in the Home App. I've sent instructions to your phone," as shown in home schematic **601**. Device **600** displays, on touchscreen display **602**, lock screen user interface **624** with setup notification **640***c*. In some embodiments, in response to selection (e.g., via a set of one or more user inputs (e.g., tap input, long-press input)) corresponding to setup notification **640***c*, device **600** displays personal request user interface **618** of FIG. **6**D. In some embodiments, in response to selection (e.g., via tap input; via long-press inputs) corresponding to setup notification **640***c*, device **600** displays an onboarding user interface similar to onboarding user interface **806** of FIG. **8**A. Initial onboarding for personal requests will be discussed further with respect to FIGS. **8**A-**8**B.

(174) FIG. **7** is a flow diagram illustrating a method for **700** using an electronic device in accordance with some embodiments. Method **700** is performed at a device (e.g., device **100**, **300**, **500**, **580**, **600**, or **603***a*-**603***d*) that is in communication with one or more input devices and a first external electronic device (e.g., device **100**, **300**, **500**, **600**). Some operations in method **700** are, optionally, combined, the orders of some operations are, optionally, changed, and some operations are, optionally, omitted.

(175) In some embodiments, the electronic device (e.g., **603***a*) is a computer system. The computer system is optionally in communication (e.g., wired communication, wireless communication) with a display generation component and with one or more input devices. The display generation component is configured to provide visual output, such as display via a CRT display, display via an LED display, or display via image projection. In some embodiments, the display generation component is integrated with the computer system. In some embodiments, the display generation component is separate from the computer system. The one or more input devices are configured to receive input, such as a touch-sensitive surface receiving user input. In some embodiments, the one or more input devices are integrated with the computer system. In some embodiments, the one or more input devices are separate from the computer system. Thus, the computer system can transmit, via a wired or wireless connection, data (e.g., image data or video data) to an integrated or external display generation component to visually produce the content (e.g., using a display device) and can receive, a wired or wireless connection, input from the one or more input devices.

(176) As described below, method **700** provides an intuitive way for managing user requests for specific operations. The method reduces the cognitive burden on a user for managing user requests

for specific operations, thereby creating a more efficient human-machine interface. For battery-operated computing devices, enabling a user to provide user requests for specific operations faster and more efficiently conserves power and increases the time between battery charges.
(177) The computer system (e.g., device **603**_a_, **603**_b_, **603**_c_, or **603**_d_) receives (**702**), via the one or more input devices (e.g., microphone **113** of device **603**_a_), a first request (e.g., **626**_a_, **628**_a_, **630**_a_, **632**_a_, **634**_a_) (e.g., a spoken request, a natural language utterance from a user of the system) of a first type (e.g., a request to perform a specific type of operation (e.g., a request to output personal data (e.g., a request for content (e.g., media; voicemail; messages; events; to-do items/lists) associated with one or more specific users (e.g., an auto-generated playlist of a user's most played songs; a user-generated playlist; serialized media (e.g., an audiobook, a podcast series); media that varies depending on the identity of the user) to perform a first operation (e.g., a data output operation that is performed using one or more output devices (e.g., an integrated audio speaker; a display) that are in communication with the computer system) (in some embodiments, the first operation is audibly outputting data corresponding to messages (e.g., voicemail, email, text messages) associated with a specific user).
(178) The computer system, in response (**704**) to receiving the first request of the first type and in accordance (**706**) with a determination that a first set of notification criteria (e.g., activity notifications setting toggled on) are met (in some embodiments, and in accordance with a determination that a first set of authorization criteria are met): performs (**708**) the first operation (e.g., outputs **626**_b_, **628**_b_, or **634**_b_); and initiates (**710**) a process to display a notification (e.g., **626**_c_, **628**_c_, **634**_c_) on the first external electronic device (e.g., **600**). In some embodiments, initiating a process to display the notification includes generating the notification and/or transmitting an indication that the first request was received to the first external electronic device or to a second external electronic device (e.g., an alert or notification server; an event-logging server in communication with a notification server) that causes display of the notification at the first external electronic device.
(179) The computer system, in response to receiving the first request of the first type and in accordance with a determination that the first set of notification criteria (e.g., activity notifications setting toggled off) are not met (in some embodiments, and in accordance with a determination that a first set of authorization criteria are met), performs (**712**) the first operation (e.g., output **626**_b_) without initiating the process to display the notification on the first external electronic device (e.g., or any other electronic device) (e.g., as seen in FIGS. **6**S and **6**T). In some embodiments, additional settings must be enabled to perform the first operation (e.g., the computer system is configured to recognize user voices; personal requests setting enabled for the requesting user). In some embodiments, a request is a request of a first type if the request meets additional criteria, such as being from a known (e.g., identified via voice recognition) user (e.g., from a set of known users). In some embodiments, the computer system receives a second request of a second type (e.g., a request for data that is not personalized; a request for data that is generic/non-variable for different users) to perform a second operation, and, in response to the second request, the computer system performs the second operation without initiating the process to display the notification on the first external electronic device (e.g., the second operation is performed without initiating the process to display the notification regardless of whether the first set of notification criteria are met). Performing the first operation with or without initiating a process to display a notification based on a set of criteria allows a user make requests of the first type, with or without the additional operation of initiating display of a notification, without requiring further inputs. In some implementations, this process seamlessly allows a user to make a request for personal information and obtain that information without additional authorization. When the process to display notifications is automatically initiated in response to requests of the first type, e.g., a personal request, privacy and security are enhanced for a user because the user is informed, on a separate device, that the computer system has provided information in response to a request. The request

and information provided may be about personal information. By notifying the user about the provided personal information, the user has a chance to review the information and subsequently manage settings in order to determine how and when to allow personal information to be given out by the computer system. Performing an optimized operation when a set of conditions are met without requiring further input and improving security and privacy aspects of interactions enhances the operability of the device and makes the user-device interface more efficient (e.g., by helping the user to provide proper inputs and reducing user mistakes when operating/interacting with the device or by restricting unauthorized inputs) which, additionally, reduces power usage and improves battery life of the device by enabling the user to use the device more quickly and efficiently.

(180) In some embodiments, the computer system is in communication with one or more output devices (e.g., speaker **111**) (e.g., an audio speaker; a display); the first request of the first type is a request to output personalized data (e.g., data corresponding to **626***b*, **628***b*, **634***b*) corresponding to a respective user associated with the request (e.g., that has been determined and/or identified (e.g., via voice recognition) as having made the first request); and performing the first operation includes: in accordance with a determination that the respective user associated with the first request is a first user (e.g., **610***a*) (e.g., a user associated with a first user account that corresponds to a first set of user-specific data), outputting, via the one or more output devices, a representation (e.g., audio outputs **626***b*, **628***b*, **634***b*) (e.g., an audio output; graphical output; a textual output) of first data that is associated with the first user; and in accordance with a determination that the respective user associated with the first request is a second user (e.g., **610***b*) (e.g., a user associated with a second user account that corresponds to a second set of user-specific data), different from the first user, outputting, via the one or more output devices, a representation (e.g., an audio output; graphical output; a textual output) of second data (e.g., and that is not associated with the first user), different from the first data, that is associated with the second user (in some embodiments, without outputting the first data). Conditionally initiating a process to display a notification based on a set of criteria in response to a request for personalized data that varies by user provides users with additional control over when such notifications are provided, without having to provide additional inputs with each request. Moreover, when the process to display notifications is automatically initiated in response to requests of the first type (e.g., personal requests), security and privacy is enhanced by informing the user, on a separate device, that a request of the first type has been processed at the computer system. By notifying the user about the provided personal information, the user has a chance to review the information and subsequently manage settings in order to determine how and when to allow personal information to be given out by the computer system. Performing an optimized operation when a set of conditions are met without requiring further input, providing users with more control, improving security, and enhancing privacy all enhance the operability of the device and makes the user-device interface more efficient (e.g., by helping the user to provide proper inputs and reducing user mistakes when operating/interacting with the device or by restricting unauthorized inputs) which, additionally, reduces power usage and improves battery life of the device by enabling the user to use the device more quickly and efficiently.

(181) In some embodiments, the first data is selected from the group consisting of an email associated with the first user, a text message associated with the first user (e.g., **626***b*), a voicemail associated with the first user, a calendar event associated with the first user (e.g., **628***b*), media (e.g., a personalized playlist) associated with the first user, a reminder associated with the first user (e.g., **634***b*), and a combination thereof.

(182) In some embodiments, the first set of notification criteria includes a criterion that is met when (in some embodiments, the first set of notification criteria are met when) a first notification setting (e.g., as seen in **605***a*) associated with the first external electronic device (e.g., associated with a user of the first external electronic device; associated with a user account that is associated

with the first external device) is enabled. In some embodiments, the first set of notification criteria are not met when the first notification setting is disabled. In some embodiments, the first external electronic device is associated with a first user account that is also associated with the computer system and the first request is determined to have been made by a user associated with the first user account. Conditionally initiating a process to display a notification based on a user-selectable notification setting provides users with additional control over when such notifications are provided, without having to provide additional inputs with each request. Moreover, doing so provides the user with the option to balance security and privacy with notification frequency. By notifying the user about the provided personal information, the user has a chance to review the information and subsequently manage settings in order to determine how and when to allow personal information to be given out by the computer system. Performing an optimized operation when set of conditions are met without requiring further input, providing users with more control, enhancing privacy, and improving security all enhance the operability of the device and makes the user-device interface more efficient (e.g., by helping the user to provide proper inputs and reducing user mistakes when operating/interacting with the device or by restricting unauthorized inputs) which, additionally, reduces power usage and improves battery life of the device by enabling the user to use the device more quickly and efficiently.
(183) In some embodiments, the first notification setting affects (e.g., controls; permits or prohibits) whether a plurality of computer systems (e.g., **603***a*, **603***b*) (e.g., computer systems configured to perform the first operation in response to requests of the first type; computer systems sharing the same hardware and/or software configuration and/or capabilities) initiate a process to display a notification on the first external device in response to receiving requests of the first type, wherein the plurality of computer systems includes the computer system. Conditionally initiating a process to display a notification based on a user-selectable notification setting that affects multiple devices provides users with additional control over when such notifications are provided for multiple devices, without having to provide additional inputs for each device and/or with each request. Moreover, doing so provides the user with the option to balance security and privacy with notification frequency for multiple devices. By notifying the user about the provided personal information, the user has a chance to review the information and subsequently manage settings in order to determine how and when to allow personal information to be given out by the computer system. Performing an optimized operation when set of conditions are met without requiring further input, providing users with more control, enhancing privacy, and improving security all enhance the operability of the device and makes the user-device interface more efficient (e.g., by helping the user to provide proper inputs and reducing user mistakes when operating/interacting with the device or by restricting unauthorized inputs) which, additionally, reduces power usage and improves battery life of the device by enabling the user to use the device more quickly and efficiently.
(184) In some embodiments, the first notification setting was enabled via a user input (e.g., an input received while displaying a settings user interface for managing the output of requests for personalized data associated with the user of the first external electronic device) received at the first external electronic device (e.g., at device **600** as seen in FIG. **6**E). Providing notifications at a specific device at which a relevant setting was enabled provides improved user feedback by increasing the likelihood that the user will notice the notification and improves security and privacy, when then notification may indicate unauthorized access or unintentional disclosure of private information. Providing improved visual feedback to the user, enhancing privacy, and improving security all enhance the operability of the device and makes the user-device interface more efficient (e.g., by helping the user to provide proper inputs and reducing user mistakes when operating/interacting with the device) which, additionally, reduces power usage and improves battery life of the device by enabling the user to use the device more quickly and efficiently.
(185) In some embodiments, the first electronic device (e.g., **600**) is configured to: in accordance

with a determination (e.g., a determination made at the computer a system; a determination made at a notification server in communication with the computer system) that a first set of notification-handling criteria are met, wherein the first set of notification-handling criteria includes a criterion that is met when a first previous notification (e.g., **626***c*) corresponding to the computer system (e.g., the first set of notification-handling criteria are met when the first previous notification remains unread and/or has not been dismissed at the first external electronic device) is currently available for display (e.g., currently being displayed), configure the notification (e.g., **628***c*) for display (e.g., displaying the notification or setting the notification for display in response to a subsequent request to display notifications) in a group (e.g., the group of FIG. **6**J) with the first previous notification (e.g., as a stack; as a single, graphical notification with an indication that the notification is associated with a plurality of notification events (e.g., a badged notification)); and in accordance with a determination (e.g., a determination made at the computer a system; a determination made at a notification server in communication with the computer system) that a second set of notification-handling criteria are met, wherein the second set of notification-handling criteria includes a criterion that is met when a second previous notification (e.g., **626***c*, **628***c*) corresponding to a second computer system (e.g., initiated in response to a second request of the first type that was received at the second computer system), different than the computer system, is currently available for display, configure the notification (e.g., **634***c*) to be displayed separately from the second previous notification (e.g., not in a group; not in a stack; as a discrete, separate notification that is displayed concurrently with the second previous notification (e.g., if the second previous notification remains unread)) (e.g., as seen in FIG. **6**P). Conditionally displaying a notification as part of a group or separately, based on a set of criteria, provides improved feedback to users as to the relationship of the notification to existing notifications. Providing improved visual feedback to the user enhances the operability of the device and makes the user-device interface more efficient (e.g., by helping the user to provide proper inputs and reducing user mistakes when operating/interacting with the device) which, additionally, reduces power usage and improves battery life of the device by enabling the user to use the device more quickly and efficiently.

(186) In some embodiments, the first electronic device is configured to: display the notification on a display (e.g., **602**) of the first electronic device; receive, via an input device (e.g., a touch sensitive surface of display **602**) (e.g., a touch-sensitive surface) of the first electronic device, a set of one or more inputs that includes an input corresponding to the notification (e.g., **636**); and in response to receiving the set of one or more inputs that includes an input corresponding to the notification (e.g., a tap, a press-and-hold gesture), display a configuration user interface that includes a first selectable user interface object (e.g., **618***a*, **618***b*, **618***c*) that, when selected, modifies a setting that affects (e.g., disables and/or enables the handling of requests of the first type; affects whether the first set of notification criteria are met in response to receiving requests of the first type) processing of requests of the first type at the computer system. Providing the user with the ability to affect settings pertaining to requests of a type that generated the notification, from the notification, gives the user improved control options and reduces the number of inputs required to access the relevant settings. Providing improved control options enhances the operability of the device and makes the user-device interface more efficient (e.g., by helping the user to provide proper inputs and reducing user mistakes when operating/interacting with the device) which, additionally, reduces power usage and improves battery life of the device by enabling the user to use the device more quickly and efficiently.

(187) In some embodiments, when the notification is grouped with one or more additional notifications, an input directed to the group of notifications causes the group to separate into discrete notifications, including displaying the notification as a discrete notification, separate from the one or more additional notifications).

(188) In some embodiments, the first electronic device is configured to display the notification as a grouped notification (e.g., group of **626***c* and **628***c* of FIG. **6**M) that corresponds to the notification

and one or more additional notifications that correspond to one or more additional requests of the first type that were received by the computer system.
(189) In some embodiments, the notification (e.g., **626***c*), when displayed at the first external electronic device, includes an indication (e.g., a textual indication; a graphical indication) (e.g., "request for messages" in **626***c*) that identifies one or more characteristics of the first operation (e.g., a type of data/content (e.g., media; voicemail; messages; events; to-do items/lists) outputted by the first operation; an application that performed the first operation). Including information that identifies one or more characteristics of the operation that was performed in response to the request provides the user with improved feedback as to the nature of the request and the operation that was performed. Providing improved visual feedback to the user enhances the operability of the device and makes the user-device interface more efficient (e.g., by helping the user to provide proper inputs and reducing user mistakes when operating/interacting with the device) which, additionally, reduces power usage and improves battery life of the device by enabling the user to use the device more quickly and efficiently.
(190) In some embodiments, the computer system, in response to receiving the first request of the first type and in accordance (**714**) with a determination that a first set of authorization criteria are not met (e.g., as seen in FIGS. **6**M and **6**N): forgoes performance (**716**) of the first operation; and outputs (**718**) an indication (e.g., **632***b*) (e.g., an audio indication; a graphical indication; a textual indication) of one or more steps that can be taken to satisfy the first set of authorization criteria for subsequent requests for the first type received at the computer system (e.g., "enable personal requests on your smart phone"). Providing the user with guidance on what steps can be taken to permit performance of the operation provides the user with feedback that the computer system is not currently configured to perform the requested operations and promotes further user-system interactions. Providing improved feedback and promoting further interactions enhances the operability of the device and makes the user-device interface more efficient (e.g., by helping the user to provide proper inputs and reducing user mistakes when operating/interacting with the device) which, additionally, reduces power usage and improves battery life of the device by enabling the user to use the device more quickly and efficiently.
(191) Note that details of the processes described above with respect to method **700** (e.g., FIG. **7**) are also applicable in an analogous manner to the methods described below. For example, method **900** optionally includes one or more of the characteristics of the various methods described above with reference to method **700**. For example, method **900** can be used to enable personal data requests that are handled according to method **700**. For brevity, these details are not repeated below.
(192) FIGS. **8**A-**8**D illustrate exemplary user interfaces for configuring the management of user requests for specific operations, in accordance with some embodiments. The user interfaces in these figures are used to illustrate the processes described below, including the processes in FIG. **9**.
(193) FIGS. **8**A-**8**D illustrate exemplary user interfaces displayed during onboarding of user accounts to use personal requests on devices (e.g., smart speaker) having a digital assistant. FIGS. **8**A-**8**C illustrate device **600**, John's phone, as discussed above. FIG. **8**D illustrates device **800**, which is user Jane Appleseed's phone. In some embodiments, device **800** includes one or more features of devices **100**, **300**, **500**, and/or **600**.
(194) In FIG. **8**A, device **600** (e.g., John's phone) displays, on touchscreen display **602**, time indication **804***a* (showing the current time of 8:00 (e.g., AM)) and onboarding user interface **806**. In some embodiments, device **600** displays onboarding user interface **806** in response to receiving a user input corresponding to selection of setup notification **640***c* of FIG. **6**V. In some embodiments, device **600** displays a set of one or more user interfaces, including onboarding user interface **806**, upon accessing the Home application (e.g., the application depicted by home user interface **606** of FIG. **6**A) for the first time after completing a software update (e.g., for device **600**, for an external device (e.g., smart speaker)). In some embodiments, device **600** displays a set of one or more user

interfaces, including onboarding user interface **806**, during setup of a new external device (e.g., smart speaker). In some embodiments, onboarding user interface **806** is shown on most of touchscreen display **602**. In some embodiments, onboarding user interface **806** is shown on a portion of touchscreen display **602**.

(195) In the scenario illustrated by FIGS. **8**A-**8**B, onboarding user interface **806** is displayed prior to the scenarios described with respect to FIGS. **6**A-**6**V. As shown in FIG. **8**A, onboarding user interface **806** generally includes example personal requests (e.g., message request **810**), example responses (e.g., calendar response **808***a*; reminders response **812***a*; call log response **814***a*), enable affordance **816***a*, and cancel affordance **816***b*. Example personal requests and example responses are dynamic examples using data specific to the user based on when onboarding user interface **806** is surfaced on device **600**. In some embodiments, the example personal requests and example responses are from any application storing personal user data (e.g., call log, voicemail, email, chat applications (e.g., messages), calendar, reminders, notes, banking (e.g., wallet), health, contacts). In some embodiments, the example requests and requests provide the user with feedback as to what personal data can be requested and what personalized responses can be provided in response to such requests.

(196) In FIG. **8**A, device **600** displays onboarding user interface **806** for enabling personal requests on John's user account at 8:00 (e.g., AM), as shown by time indication **804***a*. Calendar response **808***a* reads, “Your dentist appointment is at 12:00 PM,” which aligns with the scenario illustrated by FIGS. **6**I-**6**J, when John is asking for his next appointment at 10:10 (e.g., AM) and device **603***a* provides output, “Your dentist appointment is at 12:00 PM.” Reminder response **812***a* reads, “You have an upcoming reminder: grab milk, eggs, and sugar,” which aligns with the scenario illustrated by FIGS. **6**O-**6**P, when John is asking for his grocery list at 10:11 (e.g., AM) and device **603***b* provides output, “Your grocery list has milk, eggs, and sugar.” Call log response **814***a* reads, “You have a missed call from Jane at 7:45 AM,” which is data generated for John's user account 15 minutes prior to device **600** displaying onboarding user interface **806**. Onboarding user interface **806** also includes message request **810**, “Hey Assistant, read my last message,” as an example personal request for accessing a user's text messages. In some embodiments, any combination of example requests and example responses are shown on onboarding user interface **806**. In some embodiments, device **600** uses a generic (e.g., canned; non-specific to a user) example in place of a personalized example if the user account does not have user-specific data for an application (e.g., for a user who does not use the reminders application, show an example reminder response of “You have a reminder to pay bills.”)

(197) In FIG. **8**A, device **600** detects tap input **818** corresponding to selection of enable affordance **816***a*. As shown in FIG. **8**B, in response to tap input **818**, device **600** enables personal requests for all electronic devices (e.g., smart speakers) connected to the home network and activity notifications for John's user account and displays personal request user interface **618** associated with John's user account on touchscreen display **602**. As discussed in detail with respect to FIGS. **6**D-**6**F, personal request user interface **618** includes speaker affordance **618***a* (e.g., “BEDROOM SPEAKER”), speaker affordance **618***b* (e.g., “DINING ROOM SPEAKER”), and speaker affordance **618***c* (e.g., “LIVING ROOM SPEAKERS”) with toggles that are in the “ON” position, indicating that the electronic devices (e.g., smart speakers) associated with speaker affordances **618***a*-**618***c* are able to perform (e.g., process) personal requests for John (e.g., when the electronic device recognizes that a speech input was from John (e.g., via voice recognition)). Personal request user interface **618** of FIG. **8**B includes activity notification affordance **618***d* with a toggle that is in the “ON” position, indicating that device **600** will receive notifications whenever a device corresponding to speaker affordances **618***a*-**618***c* perform personal requests using John's user account. In some embodiments, in response to additional user inputs, device **600** shows a customized personal request configuration for John's user account, similar to those shown in FIG. **6**E and FIG. **6**F. In some embodiments, in response to selection of enable affordance **816***a*, instead

of displaying personal request user interface **618** with all toggles in the "ON" position, device **600** displays personal request user interface **618** as shown in FIG. **6**D, with all toggles in the "OFF" position.
(198) Turning back to FIG. **8**A, in some embodiments, instead of receiving a tap input at enable affordance **816***a*, device **600** detects a tap input at cancel affordance **816***b* and, in response, device **600** ceases display of onboarding user interface **806**. After 8 hours, in FIG. **8**C, device **600** redisplays on touchscreen display **602** (e.g., in response to John selection of setup notification **640***c* of FIG. **6**V at the later time), onboarding user interface **806** for enabling personal requests on John's user account at **4**:**00** (e.g., PM), as shown by time indication **804***b*. Onboarding user interface **806** of FIG. **8**C now includes different example responses (e.g., calendar response **808***b*; reminders response **812***b*; call log response **814***b*) from those in FIG. **8**A, which are now outdated.
(199) Now that the time is currently 4:00 PM and John's 12:00 PM dentist appointment already occurred, calendar response **808***b* reads, "Your next appointment is at 5:00 PM." In some embodiments, calendar response **808***b* is another appointment scheduled for John's user account. In some embodiments, calendar response **808***b* is a canned example for a calendar response if John's user account does not include additional appointments. Similarly, John completed his grocery shopping for milk, eggs, and sugar and updated his reminders application, so reminder response **812***b* reads, "You have one reminder for today: Pay power bill." In some embodiments, reminder response **812***b* is another reminder listed for John's user account. In some embodiments, reminder response **812***b* is a canned example for a reminder response if John's user account does not include additional reminders. As the day progresses, device **600** receives additional phone calls for John, and now, call log response **814***b* reads, "You have two missed calls from Grandpa Appleseed." Onboarding user interface **806** includes message request **810**, "Hey Assistant, read my last message," which is unchanged from onboarding user interface **806** of FIG. **8**A. In some embodiments, example personal requests and example responses include varying levels of detail (e.g., "You have a missed call from Jane at 7:45 AM;" "You have two missed calls from Grandpa Appleseed;" "You have two missed calls"). As seen in FIG. **8**C, the example requests and responses can dynamically update, depending on the current personalized data associated with user John.
(200) In some embodiments, one or more example personal requests and example responses are the same between two instances of accessing the onboarding user interface **806**. In some embodiments, one or more example personal requests and example responses change between two instances of accessing the onboarding user interface **806** based on a period of time (e.g., minutes, hours, days). In some embodiments, one or more example personal requests and example responses change between two instances of accessing the onboarding user interface **806** based on user data (e.g., completion of a reminder; received calls; received messages).
(201) In addition to dynamically changing over time and based on user data for a particular user, onboarding user interface **806** can vary between different users, based on their different personalized data. In FIG. **8**D, device **800** (e.g., Jane's phone) displays, on touchscreen display **802**, onboarding user interface **806** for enabling personal requests on Jane's user account at **4**:**00** (e.g., PM), as shown by time indication **804***b*. Onboarding user interface **806** of FIG. **8**D includes example personal requests (e.g., message request **810**), example responses (e.g., calendar response **808***c*; reminders response **812***c*; call log response **814***c*), enable affordance **816***a*, and cancel affordance **816***b*. Message request **810** is unchanged between onboarding user interface **806** for Jane's user account and onboarding user interface **806** of FIGS. **8**A and **8**C for John's user account. In some embodiments, example personal requests or example responses are the same between different users. Calendar response **808***c*, reminders response **812***c*, call log response **814***c* corresponding to Jane's user account are all different from those shown in FIGS. **8**A and **8**C for John's user account.
(202) FIG. **9** is a flow diagram illustrating a method for configuring management of user requests for specific operations using an electronic device in accordance with some embodiments. Method

900 is performed at a device (e.g., **100**, **300**, **500**, **600**) with a display generation component (e.g., **602**). Some operations in method **900** are, optionally, combined, the orders of some operations are, optionally, changed, and some operations are, optionally, omitted.

(203) In some embodiments, the electronic device (e.g., **600**) is a computer system. The computer system is optionally in communication (e.g., wired communication, wireless communication) with a display generation component and with one or more input devices. The display generation component is configured to provide visual output, such as display via a CRT display, display via an LED display, or display via image projection. In some embodiments, the display generation component is integrated with the computer system. In some embodiments, the display generation component is separate from the computer system. The one or more input devices are configured to receive input, such as a touch-sensitive surface receiving user input. In some embodiments, the one or more input devices are integrated with the computer system. In some embodiments, the one or more input devices are separate from the computer system. Thus, the computer system can transmit, via a wired or wireless connection, data (e.g., image data or video data) to an integrated or external display generation component to visually produce the content (e.g., using a display device) and can receive, a wired or wireless connection, input from the one or more input devices.

(204) As described below, method **900** provides an intuitive way for configuring management of user requests for specific operations. The method reduces the cognitive burden on a user for configuring management of user requests for specific operations, thereby creating a more efficient human-machine interface. For battery-operated computing devices, enabling a user to configure management of user requests for specific operations faster and more efficiently conserves power and increases the time between battery charges.

(205) The computer system (e.g., **600**) displays (**902**), via the display generation component (e.g., **602**), a first configuration user interface (e.g., **806**) associated with a process for configuring data request operations (e.g., a request to perform a specific type of operation (e.g., a request to output personal data (e.g., a request for content (e.g., media; voicemail; messages; events; to-do items/lists) associated with one or more specific users (e.g., an auto-generated playlist of a user's most played songs; a user-generated playlist; serialized media (e.g., an audiobook, a podcast series); media that varies depending on the identity of the user) for a respective user (e.g., John Appleseed), wherein the first configuration user interface includes a set of one or more indications (e.g., **808***a*, **810**, **812***a*, **814***a*) (e.g., indications of exemplary operation request inputs that can be made; indications of exemplary outputs that are provided in response to data request operations; graphical and/or textual indications), personalized for the respective user, that includes a first indication (e.g., **808***b*, **808***c*), wherein: in accordance (**904**) with a determination that the respective user is a first user (e.g., John Appleseed) (e.g., a user associated with a first user account that corresponds to a first set of user-specific data), the first indication includes a representation (e.g., **808***b*) (e.g., a graphical representation; a textual representation) of first data (e.g., data corresponding to an upcoming calendar appointment for the first user (e.g., “Team dinner at 6:00 PM”); recent call log; recently received messages; outstanding reminders (e.g., “Pick up dry cleaning”)) that is associated with the first user; and in accordance (**906**) with a determination that the respective user is a second user (e.g., Jane Appleseed) (e.g., a user associated with a second user account, different from the first user account, that corresponds to a second set of user-specific data), different from the first user, the first indication includes a representation (e.g., **808***c*) of second data, different from the first data, that is associated with the second user (e.g., and that is not associated with the first user). In some embodiments, in accordance with the respective user being a second user, the first indication (in some embodiments, the first configuration user interface as a whole) does not include the representation of the first data. Providing the user with personalized/customized indications provides the user with feedback as to what data the system has access to and what requests can be made to the system and/or what responses can result and also promotes further user-system interactions. These personalized examples can alert the user to the

private information that may be shared by the computer system. By viewing personal examples of real information that may be shared, the user can make an informed decision about whether the user wants to share the personal information or protect the information from being disclosed by the system. This process therefore allows the user to maintain privacy and gives the user more control over personal information. Providing improved feedback and promoting further interactions enhances the operability of the device and makes the user-device interface more efficient (e.g., by helping the user to provide proper inputs and reducing user mistakes when operating/interacting with the device) which, additionally, reduces power usage and improves battery life of the device by enabling the user to use the device more quickly and efficiently.

(206) In some embodiments, the representation of the first data is: in accordance with a determination that the first configuration user interface is displayed at a first time (e.g., 8 AM as seen in FIG. **8**A) (e.g., initially displayed at the first time; displayed in response to a request received at the first time), a representation of data (e.g., **808***a*) associated with the first user that is relevant (e.g., based on the first time; determined and/or identified using the first time) to the first time (e.g., an upcoming (e.g., next) appointment that is closest in time to the first time); and in accordance with a determination that the first configuration user interface is displayed at a second time (e.g., 4 PM), different than the first time (e.g. later than the first time (e.g., 6 PM with the first time being 3 PM), a representation of data (e.g., **808***b*) associated with the first user that is relevant to the second time (e.g., that is not relevant to the first time) (in some embodiments, without displaying the representation of data associated with the first user that is relevant to the first time), wherein the representation of data associated with the first user that is relevant to the first time is different than the representation of data associated with the first user that is relevant to the second time. In some embodiments, the first configuration user interface includes different indications for a user, depending on the time at which the interface is displayed. Providing the user with time-relevant indications provides the user with feedback as to the current time and as to what data the system has access to at the current time and also promotes further user-system interactions. Providing improved feedback and promoting further interactions enhances the operability of the device and makes the user-device interface more efficient (e.g., by helping the user to provide proper inputs and reducing user mistakes when operating/interacting with the device) which, additionally, reduces power usage and improves battery life of the device by enabling the user to use the device more quickly and efficiently.

(207) In some embodiments, the representation of the first data includes: in accordance with a determination that the first user is associated with a first set of personalized data (e.g., media; voicemails; messages; calendar events; to-do items/lists) that includes third data (e.g., data corresponding to **808***a*) (e.g., a first calendar event), a representation of the third data (e.g., **808***a*); and in accordance with a determination that the first user is associated with a second set of personalized data (e.g., data corresponding to **808***b*) that is different than the first set of personalized data and that includes fourth data (e.g., a second calendar event, different from the first calendar event) (in some embodiments, the second set of personalized data does not include the third data), a representation of the fourth data (e.g., **808***b*). In some embodiments, the first configuration user interface includes different indications for a user, depending on the personalized data associated with that user. Providing the user with indications that are personalized to the user's current data set provides the user with feedback as to what data the system has access to and also promotes further user-system interactions. Providing improved feedback and promoting further interactions enhances the operability of the device and makes the user-device interface more efficient (e.g., by helping the user to provide proper inputs and reducing user mistakes when operating/interacting with the device) which, additionally, reduces power usage and improves battery life of the device by enabling the user to use the device more quickly and efficiently.

(208) In some embodiments, the first indication corresponds to a first category of indications (e.g., **808***a* (calendar events) (e.g., indications corresponding to calendar events); and the set of one or

more indications includes a second indication (e.g., **812***a* (reminders)), corresponding to a second category of indications, different from the first category of indications (e.g., indications corresponding to reminders), that: in accordance with a determination that the respective user is associated with personalized data of the second category, includes a representation of fifth data (e.g., **812***a*) that is associated with the respective user (e.g., that is specific to the respective user; that is identified from a set of personalized data associated with the respective user); and in accordance with a determination that the respective user is not associated with personalized data of the second category (e.g., the respective user does not have reminders), includes a representation of sixth data that is not associated with the respective user (e.g., is not specific to the respective user; is common to multiple/all users; is a generic indication that is displayed for any user that does not have personalized data of the second category).
(209) In some embodiments, the first indication is a representation of a request for output (e.g., **810**) of personalized data associated with the respective user that can be provided to (e.g., provided as an input to) a second computer system (e.g., a computer system in communication with the computer system). In some embodiments, the first indication is a representation of an output that is provided in response to a request for output of personalized data associated with the respective user that can provided to a second computer system. Providing the user with indications of types of requests that can be made provides the user with feedback as to what requests the system can process and also promotes further user-system interactions. Providing improved feedback and promoting further interactions enhances the operability of the device and makes the user-device interface more efficient (e.g., by helping the user to provide proper inputs and reducing user mistakes when operating/interacting with the device) which, additionally, reduces power usage and improves battery life of the device by enabling the user to use the device more quickly and efficiently.
(210) Note that details of the processes described above with respect to method **900** (e.g., FIG. **9**) are also applicable in an analogous manner to the methods described above. For example, method **900** optionally includes one or more of the characteristics of the various methods described above with reference to method **700**. For example, Prior to handling requests according to method **700**, a device can be configured to handle requests using method **900**. For brevity, these details are not repeated below.
(211) The foregoing description, for purpose of explanation, has been described with reference to specific embodiments. However, the illustrative discussions above are not intended to be exhaustive or to limit the invention to the precise forms disclosed. Many modifications and variations are possible in view of the above teachings. The embodiments were chosen and described in order to best explain the principles of the techniques and their practical applications. Others skilled in the art are thereby enabled to best utilize the techniques and various embodiments with various modifications as are suited to the particular use contemplated.
(212) Although the disclosure and examples have been fully described with reference to the accompanying drawings, it is to be noted that various changes and modifications will become apparent to those skilled in the art. Such changes and modifications are to be understood as being included within the scope of the disclosure and examples as defined by the claims.
(213) As described above, one aspect of the present technology is the gathering and use of data available from various sources to improve the handling of user requests for specific operations. The present disclosure contemplates that in some instances, this gathered data may include personal information data that uniquely identifies or can be used to contact or locate a specific person. Such personal information data can include demographic data, location-based data, telephone numbers, email addresses, twitter IDs, home addresses, data or records relating to a user's health or level of fitness (e.g., vital signs measurements, medication information, exercise information), date of birth, or any other identifying or personal information.
(214) The present disclosure recognizes that the use of such personal information data, in the

present technology, can be used to the benefit of users. For example, the personal information data can be used to improve the handling of user requests for specific operations. Accordingly, use of such personal information data enables users to have calculated control of the handling of user requests for specific operations. Further, other uses for personal information data that benefit the user are also contemplated by the present disclosure. For instance, health and fitness data may be used to provide insights into a user's general wellness, or may be used as positive feedback to individuals using technology to pursue wellness goals.
(215) The present disclosure contemplates that the entities responsible for the collection, analysis, disclosure, transfer, storage, or other use of such personal information data will comply with well-established privacy policies and/or privacy practices. In particular, such entities should implement and consistently use privacy policies and practices that are generally recognized as meeting or exceeding industry or governmental requirements for maintaining personal information data private and secure. Such policies should be easily accessible by users, and should be updated as the collection and/or use of data changes. Personal information from users should be collected for legitimate and reasonable uses of the entity and not shared or sold outside of those legitimate uses. Further, such collection/sharing should occur after receiving the informed consent of the users. Additionally, such entities should consider taking any needed steps for safeguarding and securing access to such personal information data and ensuring that others with access to the personal information data adhere to their privacy policies and procedures. Further, such entities can subject themselves to evaluation by third parties to certify their adherence to widely accepted privacy policies and practices. In addition, policies and practices should be adapted for the particular types of personal information data being collected and/or accessed and adapted to applicable laws and standards, including jurisdiction-specific considerations. For instance, in the US, collection of or access to certain health data may be governed by federal and/or state laws, such as the Health Insurance Portability and Accountability Act (HIPAA); whereas health data in other countries may be subject to other regulations and policies and should be handled accordingly. Hence different privacy practices should be maintained for different personal data types in each country.
(216) Despite the foregoing, the present disclosure also contemplates embodiments in which users selectively block the use of, or access to, personal information data. That is, the present disclosure contemplates that hardware and/or software elements can be provided to prevent or block access to such personal information data. For example, in the case of handling of user requests for specific operations, the present technology can be configured to allow users to select to “opt in” or “opt out” of participation in the collection of personal information data during registration for services or anytime thereafter. In another example, users can select not to provide notification-preference data for targeted content delivery services. In yet another example, users can select to limit the length of time notification-preference data is maintained or entirely prohibit the development of a baseline mood profile. In addition to providing “opt in” and “opt out” options, the present disclosure contemplates providing notifications relating to the access or use of personal information. For instance, a user may be notified upon downloading an app that their personal information data will be accessed and then reminded again just before personal information data is accessed by the app.
(217) Moreover, it is the intent of the present disclosure that personal information data should be managed and handled in a way to minimize risks of unintentional or unauthorized access or use. Risk can be minimized by limiting the collection of data and deleting data once it is no longer needed. In addition, and when applicable, including in certain health related applications, data de-identification can be used to protect a user's privacy. De-identification may be facilitated, when appropriate, by removing specific identifiers (e.g., date of birth, etc.), controlling the amount or specificity of data stored (e.g., collecting location data a city level rather than at an address level), controlling how data is stored (e.g., aggregating data across users), and/or other methods.
(218) Therefore, although the present disclosure broadly covers use of personal information data to

implement one or more various disclosed embodiments, the present disclosure also contemplates that the various embodiments can also be implemented without the need for accessing such personal information data. That is, the various embodiments of the present technology are not rendered inoperable due to the lack of all or a portion of such personal information data. For example, content (e.g., notifications) can be selected and delivered to users by inferring preferences based on non-personal information data or a bare minimum amount of personal information, such as the content being requested by the device associated with a user, other non-personal information available to the notification delivery services, or publicly available information.

Claims

1. A computer system configured to be in communication with one or more input devices, one or more output devices, and a first external electronic device, comprising: one or more processors; memory storing one or more programs configured to be executed by the one or more processors, the one or more programs including instructions for: receiving, via the one or more input devices, a first request to perform a respective operation; and in response to receiving the first request: in accordance with a determination that the first request is a request to access first content associated with a first user account associated with the computer system, and a determination that a first set of notification criteria are met and a first set of request criteria are met, wherein the first set of request criteria includes a first criterion that is met when the computer system is enabled to output content associated with a respective user account: performing a first operation including outputting a first representation of data that includes the first content associated with the first user account via a first output device of the one or more output devices; and initiating a process to display a notification on the first external electronic device, wherein the notification includes an indication that the respective operation was performed; in accordance with a determination that the first request is a request to access second content associated with a second user account associated with the computer system, and a determination that the first set of notification criteria are met and the first set of request criteria are met: performing a second operation including outputting a second representation of data that includes the second content associated with the second user account via the first output device of the one or more output devices; and initiating the process to display the notification; in accordance with a determination that the first request is the request to access first content associated with the first user account associated with the computer system, and a determination that a first set of notification criteria are met and the first set of request criteria are not met: performing a third operation including outputting a third representation of data that does not include the first content associated with the first user account via a first output device of the one or more output devices; and initiating a process to display a notification on the first external electronic device; and in accordance with a determination that the first request is a request to access third content, different from the first content and the second content, that is not associated with the first user account or the second user account associated with the computer system, performing a fourth operation, different from the first operation and the second operation, including outputting the third content via the same first output device without initiating the process to display the notification, wherein performing the third operation includes outputting a representation of the third content that is non-variable for different users associated with the request.
2. The computer system of claim 1, wherein the first content associated with the first user account is selected from a group consisting of an email associated with the first user account, a text message associated with the first user account, a voicemail associated with the first user account, a calendar event associated with the first user account, media associated with the first user account, a reminder associated with the first user account, and a combination thereof.
3. The computer system of claim 1, wherein: the process to display the notification is a process for displaying the notification on the first external electronic device; and the first set of notification

criteria includes a criterion that is met when a first notification setting associated with the first external electronic device is enabled.
4. The computer system of claim 3, wherein the first notification setting affects whether a plurality of computer systems initiate a process to display a notification on the first external electronic device in response to receiving requests to access respective content associated with the first user account associated with the computer system, wherein the plurality of computer systems includes the computer system.
5. The computer system of claim 3, wherein the first notification setting was enabled via a user input received at the first external electronic device.
6. The computer system of claim 1, wherein: the process to display the notification is a process for displaying the notification on the first external electronic device; and the first external electronic device is configured to: in accordance with a determination that a first set of notification-handling criteria are met, wherein the first set of notification-handling criteria includes a criterion that is met when a first previous notification corresponding to the computer system is currently available for display, configure the notification for display in a group with the first previous notification; and in accordance with a determination that a second set of notification-handling criteria are met, wherein the second set of notification-handling criteria includes a criterion that is met when a second previous notification corresponding to a second computer system, different than the computer system, is currently available for display, configure the notification to be displayed separately from the second previous notification.
7. The computer system of claim 1, wherein: the process to display the notification is a process for displaying the notification on the first external electronic device; and the first external electronic device is configured to: display the notification on a display of the first external electronic device; receive, via an input device of the first external electronic device, a set of one or more inputs that includes an input corresponding to the notification; and in response to receiving the set of one or more inputs that includes an input corresponding to the notification, display a configuration user interface that includes a first selectable user interface object that, when selected, modifies a setting that affects processing of requests to access respective content associated with the first user account associated with the computer system.
8. The computer system of claim 7, wherein: the process to display the notification is a process for displaying the notification on the first external electronic device; and the first external electronic device is configured to display the notification as a grouped notification that corresponds to the notification and one or more additional notifications that correspond to one or more additional requests to perform operations of a first type that were received by the computer system.
9. The computer system of claim 1, wherein: the process to display the notification is a process for displaying the notification on the first external electronic device; and the notification, when displayed at the first external electronic device, includes an indication that identifies one or more characteristics of the respective operation.
10. The computer system of claim 1, the one or more programs further including instructions for: in response to receiving the first request and in accordance with the determination that the first request is the request to access the first content associated with the first user account associated with the computer system: in accordance with a determination that a first set of authorization criteria are not met: forgoing performing the first operation; and outputting an indication of one or more steps that can be taken to satisfy the first set of authorization criteria for subsequent requests to access respective content associated with the first user account associated with the computer system.
11. A non-transitory computer-readable storage medium storing one or more programs configured to be executed by one or more processors of a computer system that is in communication with one or more input devices, one or more output devices, and a first external electronic device, the one or more programs including instructions for: receiving, via the one or more input devices, a first

request to perform a respective operation; and in response to receiving the first request: in accordance with a determination that the first request is a request to access first content associated with a first user account associated with the computer system, and a determination that a first set of notification criteria are met and a first set of request criteria are met, wherein the first set of request criteria includes a first criterion that is met when the computer system is enabled to output content associated with a respective user account: performing a first operation including outputting a first representation of data that includes the first content associated with the first user account via a first output device of the one or more output devices; and initiating a process to display a notification on the first external electronic device, wherein the notification includes an indication that the respective operation was performed; in accordance with a determination that the first request is a request to access second content associated with a second user account associated with the computer system, and a determination that the first set of notification criteria are met and the first set of request criteria are met: performing a second operation including outputting a second representation of data that includes the second content associated with the second user account via the first output device of the one or more output devices; and initiating the process to display the notification; in accordance with a determination that the first request is the request to access first content associated with the first user account associated with the computer system, and a determination that a first set of notification criteria are met and the first set of request criteria are not met: performing a third operation including outputting a third representation of data that does not include the first content associated with the first user account via a first output device of the one or more output devices; and initiating a process to display a notification on the first external electronic device; and in accordance with a determination that the first request is a request to access third content, different from the first content and the second content, that is not associated with the first user account or the second user account associated with the computer system, performing a fourth operation, different from the first operation and the second operation, including outputting the third content via the same first output device without initiating the process to display the notification, wherein performing the third operation includes outputting a representation of the third content that is non-variable for different users associated with the request.

12. The non-transitory computer-readable storage medium of claim 11, wherein the first content associated with the first user account is selected from a group consisting of an email associated with the first user account, a text message associated with the first user account, a voicemail associated with the first user account, a calendar event associated with the first user account, media associated with the first user account, a reminder associated with the first user account, and a combination thereof.

13. The non-transitory computer-readable storage medium of claim 11, wherein: the process to display the notification is a process for displaying the notification on the first external electronic device; and the first set of notification criteria includes a criterion that is met when a first notification setting associated with the first external electronic device is enabled.

14. The non-transitory computer-readable storage medium of claim 13, wherein the first notification setting affects whether a plurality of computer systems initiate a process to display a notification on the first external electronic device in response to receiving requests to access respective content associated with the first user account associated with the computer system, wherein the plurality of computer systems includes the computer system.

15. The non-transitory computer-readable storage medium of claim 13, wherein the first notification setting was enabled via a user input received at the first external electronic device.

16. The non-transitory computer-readable storage medium of claim 11, wherein: the process to display the notification is a process for displaying the notification on the first external electronic device; and the first external electronic device is configured to: in accordance with a determination that a first set of notification-handling criteria are met, wherein the first set of notification-handling criteria includes a criterion that is met when a first previous notification corresponding to the

computer system is currently available for display, configure the notification for display in a group with the first previous notification; and in accordance with a determination that a second set of notification-handling criteria are met, wherein the second set of notification-handling criteria includes a criterion that is met when a second previous notification corresponding to a second computer system, different than the computer system, is currently available for display, configure the notification to be displayed separately from the second previous notification.
17. The non-transitory computer-readable storage medium of claim 11, wherein: the process to display the notification is a process for displaying the notification on the first external electronic device; and the first external electronic device is configured to: display the notification on a display of the first external electronic device; receive, via an input device of the first external electronic device, a set of one or more inputs that includes an input corresponding to the notification; and in response to receiving the set of one or more inputs that includes an input corresponding to the notification, display a configuration user interface that includes a first selectable user interface object that, when selected, modifies a setting that affects processing of requests to access respective content associated with the first user account associated with the computer system.
18. The non-transitory computer-readable storage medium of claim 17, wherein: the process to display the notification is a process for displaying the notification on the first external electronic device; and the first external electronic device is configured to display the notification as a grouped notification that corresponds to the notification and one or more additional notifications that correspond to one or more additional requests to perform operations of a first type that were received by the computer system.
19. The non-transitory computer-readable storage medium of claim 11, wherein: the process to display the notification is a process for displaying the notification on the first external electronic device; and the notification, when displayed at the first external electronic device, includes an indication that identifies one or more characteristics of respective first operation.
20. The non-transitory computer-readable storage medium of claim 11, the one or more programs further including instructions for: in response to receiving the first request and in accordance with the determination that the first request is the request to access the first content associated with the first user account associated with the computer system: in accordance with a determination that a first set of authorization criteria are not met: forgoing performing the first operation; and outputting an indication of one or more steps that can be taken to satisfy the first set of authorization criteria for subsequent requests to access respective content associated with the first user account associated with the computer system.
21. A method comprising: at a computer system that is in communication with one or more input devices, one or more output devices, and a first external electronic device: receiving, via the one or more input devices, a first request to perform a respective operation; and in response to receiving the first request: in accordance with a determination that the first request is a request to access first content associated with a first user account associated with the computer system, and a determination that a first set of notification criteria are met and a first set of request criteria are met, wherein the first set of request criteria includes a first criterion that is met when the computer system is enabled to output content associated with a respective user account: performing a first operation including outputting a first representation of data that includes the first content associated with the first user account via a first output device of the one or more output devices; and initiating a process to display a notification on the first external electronic device, wherein the notification includes an indication that the respective operation was performed; in accordance with a determination that the first request is a request to access second content associated with a second user account associated with the computer system, and a determination that the first set of notification criteria are met and the first set of request criteria are met: performing a second operation including outputting a second representation of data that includes the second content associated with the second user account via the first output device of the one or more output

devices; and initiating the process to display the notification; in accordance with a determination that the first request is the request to access first content associated with the first user account associated with the computer system, and a determination that a first set of notification criteria are met and the first set of request criteria are not met: performing a third operation including outputting a third representation of data that does not include the first content associated with the first user account via a first output device of the one or more output devices; and initiating a process to display a notification on the first external electronic device; and in accordance with a determination that the first request is a request to access third content, different from the first content and the second content, that is not associated with the first user account or the second user account associated with the computer system, performing a fourth operation, different from the first operation and the second operation, including outputting the third content via the same first output device without initiating the process to display the notification, wherein performing the third operation includes outputting a representation of the third content that is non-variable for different users associated with the request.

22. The method of claim 21, wherein the first content associated with the first user account is selected from a group consisting of an email associated with the first user account, a text message associated with the first user account, a voicemail associated with the first user account, a calendar event associated with the first user account, media associated with the first user account, a reminder associated with the first user account, and a combination thereof.

23. The method of claim 21, wherein: the process to display the notification is a process for displaying the notification on the first external electronic device; and the first set of notification criteria includes a criterion that is met when a first notification setting associated with the first external electronic device is enabled.

24. The method of claim 23, wherein the first notification setting affects whether a plurality of computer systems initiate a process to display a notification on the first external electronic device in response to receiving requests to access respective content associated with the first user account associated with the computer system, wherein the plurality of computer systems includes the computer system.

25. The method of claim 23, wherein the first notification setting was enabled via a user input received at the first external electronic device.

26. The method of claim 21, wherein: the process to display the notification is a process for displaying the notification on the first external electronic device; and the first external electronic device is configured to: in accordance with a determination that a first set of notification-handling criteria are met, wherein the first set of notification-handling criteria includes a criterion that is met when a first previous notification corresponding to the computer system is currently available for display, configure the notification for display in a group with the first previous notification; and in accordance with a determination that a second set of notification-handling criteria are met, wherein the second set of notification-handling criteria includes a criterion that is met when a second previous notification corresponding to a second computer system, different than the computer system, is currently available for display, configure the notification to be displayed separately from the second previous notification.

27. The method of claim 21, wherein: the process to display the notification is a process for displaying the notification on the first external electronic device; and the first external electronic device is configured to: display the notification on a display of the first external electronic device; receive, via an input device of the first external electronic device, a set of one or more inputs that includes an input corresponding to the notification; and in response to receiving the set of one or more inputs that includes an input corresponding to the notification, display a configuration user interface that includes a first selectable user interface object that, when selected, modifies a setting that affects processing of requests to access respective content associated with the first user account associated with the computer system.

28. The method of claim 27, wherein: the process to display the notification is a process for displaying the notification on the first external electronic device; and the first external electronic device is configured to display the notification as a grouped notification that corresponds to the notification and one or more additional notifications that correspond to one or more additional requests to perform operations of a first type that were received by the computer system.
29. The method of claim 21, wherein: the process to display the notification is a process for displaying the notification on the first external electronic device; and the notification, when displayed at the first external electronic device, includes an indication that identifies one or more characteristics of the respective operation.
30. The method of claim 21, further comprising: in response to receiving the first request and in accordance with the determination that the first request is the request to access the first content associated with the first user account associated with the computer system: in accordance with a determination that a first set of authorization criteria are not met: forgoing performing the first operation; and outputting an indication of one or more steps that can be taken to satisfy the first set of authorization criteria for subsequent requests to access respective content associated with the first user account associated with the computer system.